FARMMANAGEMENTSYSTEM

UserStory:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I wanttouseacentralizedSalesforce-basedplatformtorecordandmanage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validationrulesto prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive informationisprotectedwhileenablingsmoothcollaborationamong users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

ProjectOverview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

Purpose:

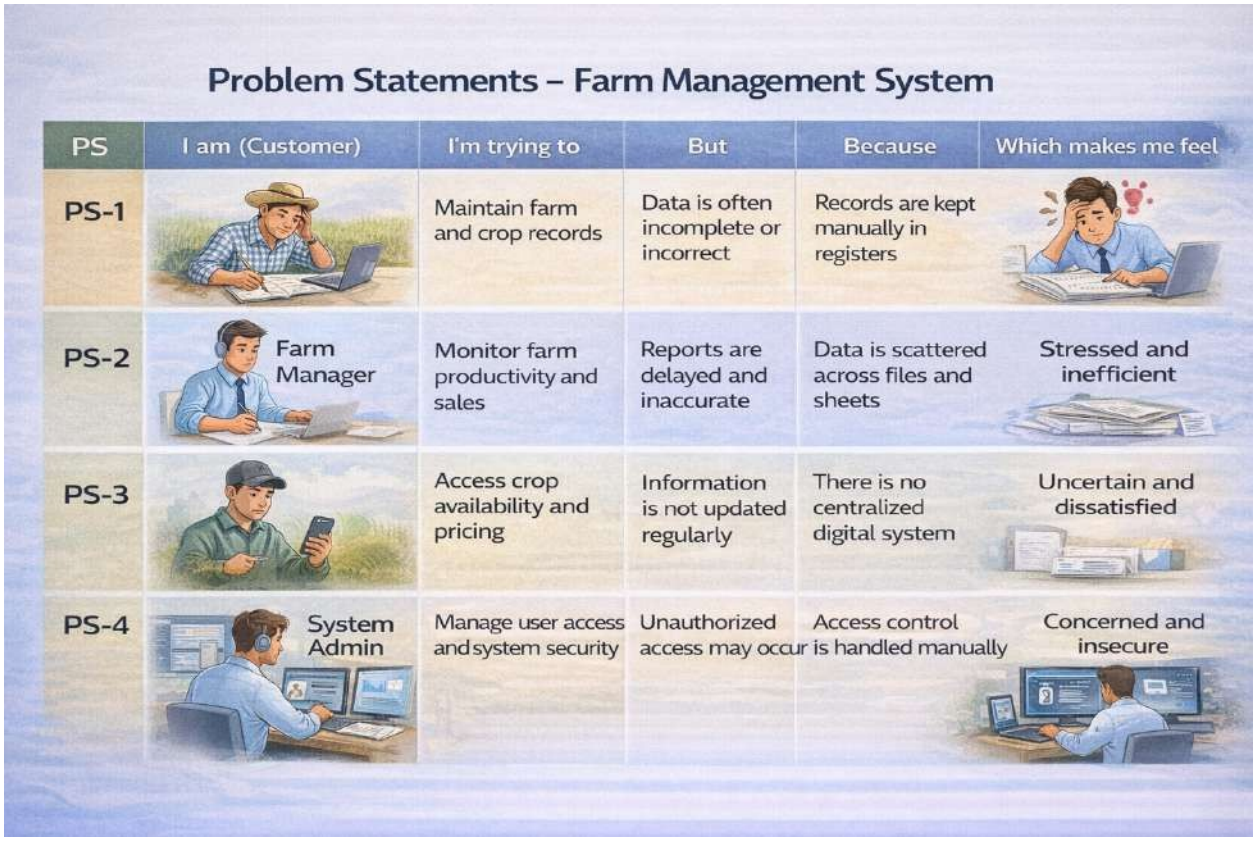
Themainpurposeofthisprojectisto:

- Digitizeagriculturaldatamanagement
- Improvedataaccuracyandaccessibility
- Automaterecordhandling
- Enablereal-timemonitoring
- Supportdata-drivenfarmingdecisions
- Enhanceproductivityandtransparency

2. IDEATIONPHASE

ProblemStatement(CustomerProblemStatementTemplate)

PS	I am (Customer)	I'mtrying to	But	Because	Which makesme feel
PS-1	Farmer	Maintain farm and croprecords	Dataisoften incomplete or incorrect	Recordsare kept manuallyin registers	Frustrated and confused
PS-2	FarmManager	Monitor farm productivity and sales	Reports are delayedand inaccurate	Data is scattered acrossfiles and sheets	Stressedand inefficient
PS-3	Buyer	Accesscrop availability and pricing	Information is not updated regularly	Thereisno centralized digital system	Uncertain and dissatisfied
PS-4	SystemAdmin	Manageuser access and system security	Unauthorized access may occur	Access controlis handled manually	Concerned andinsecure



Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

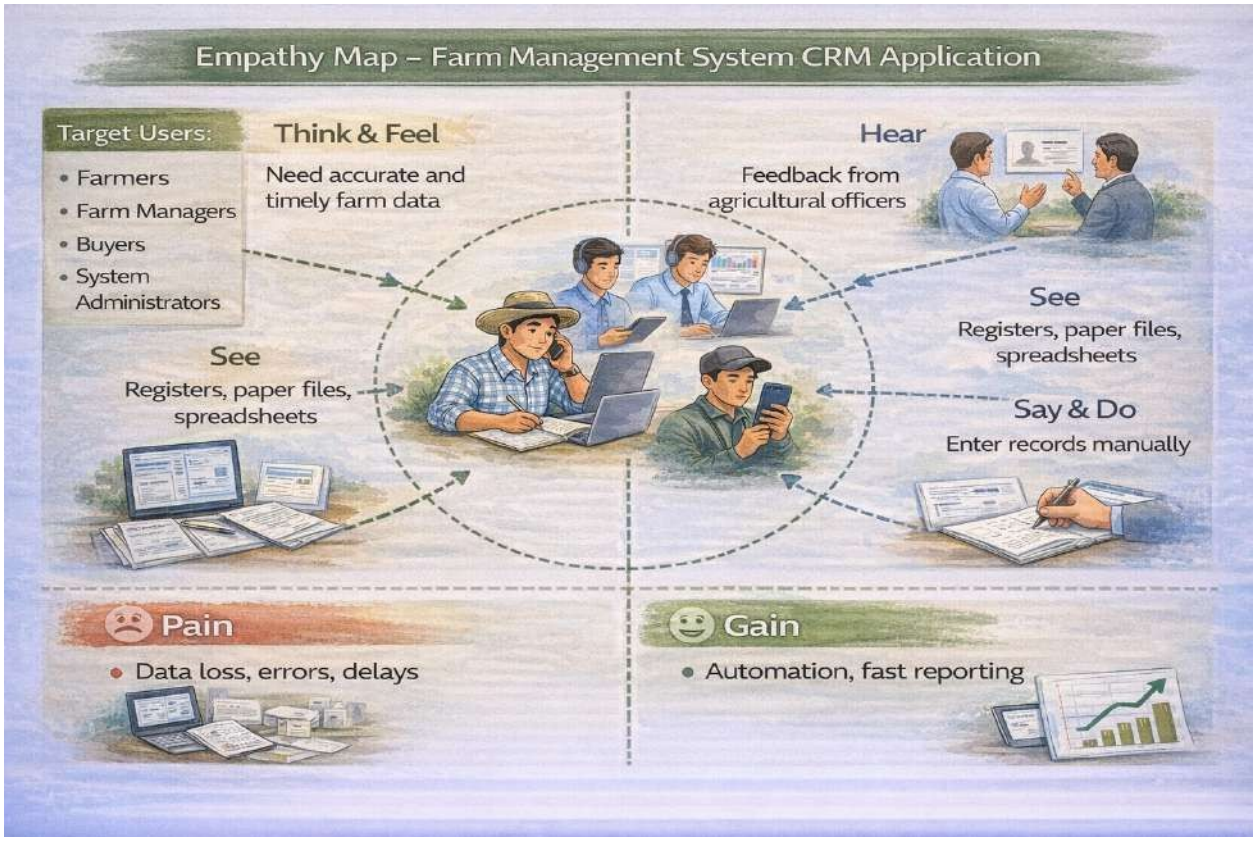
EmpathyMapCanvas

TargetUsers:

- Farmers
- FarmManagers
- Buyers
- SystemAdministrators

EmpathyAnalysis

- **Think&Feel:**Needaccurateandtimelyfarm data
- **Hear:**Feedbackfromagriculturalofficers
- **See:**Registers,paperfiles,spreadsheets
- **Say&Do:**Enterrecordsmanually
- **Pain:**Dataloss,errors,delays
- **Gain:**Automation,fastreporting



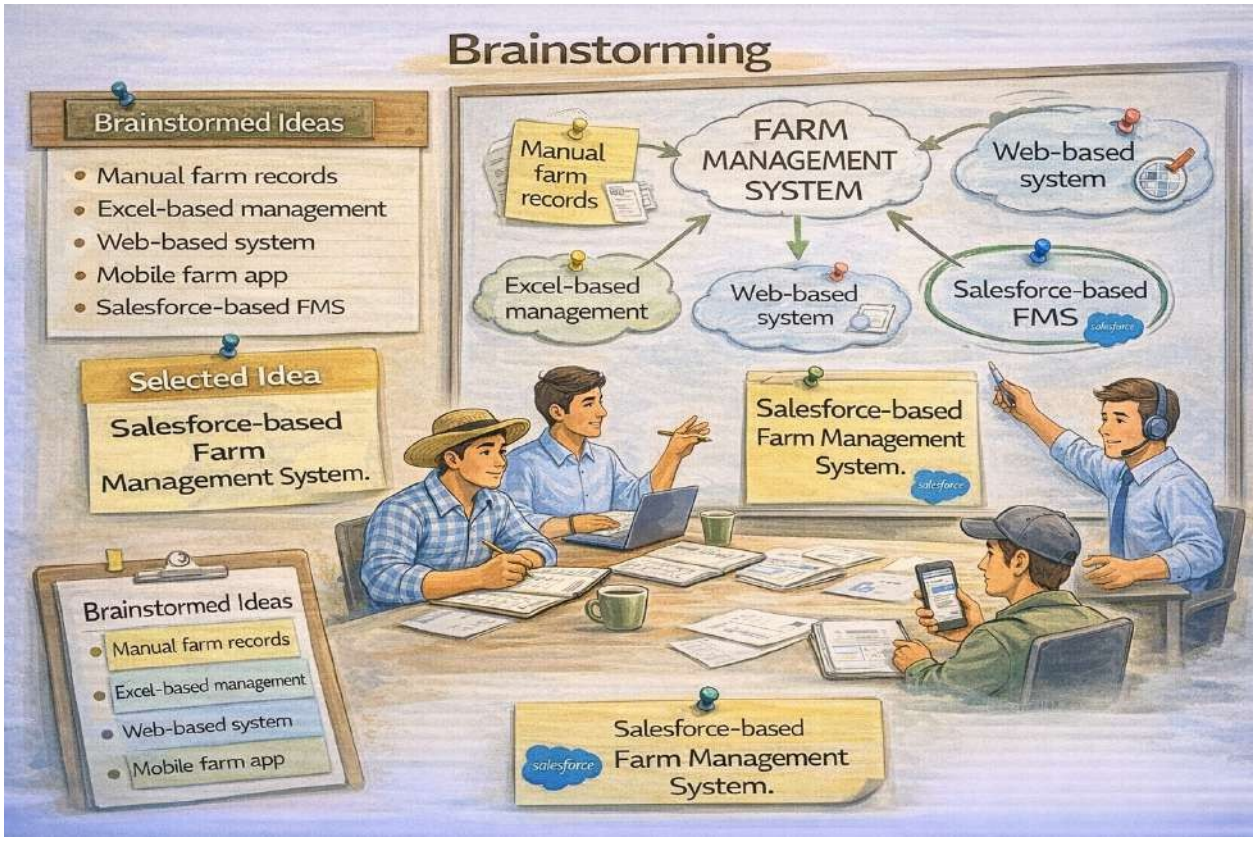
Brainstorming

BrainstormedIdeas

- Manualfarmrecords
- Excel-basedmanagement
- Web-basedsystem
- Mobilefarm app
- Salesforce-basedFMS

SelectedIdea

Salesforce-basedFarmManagementSystem.

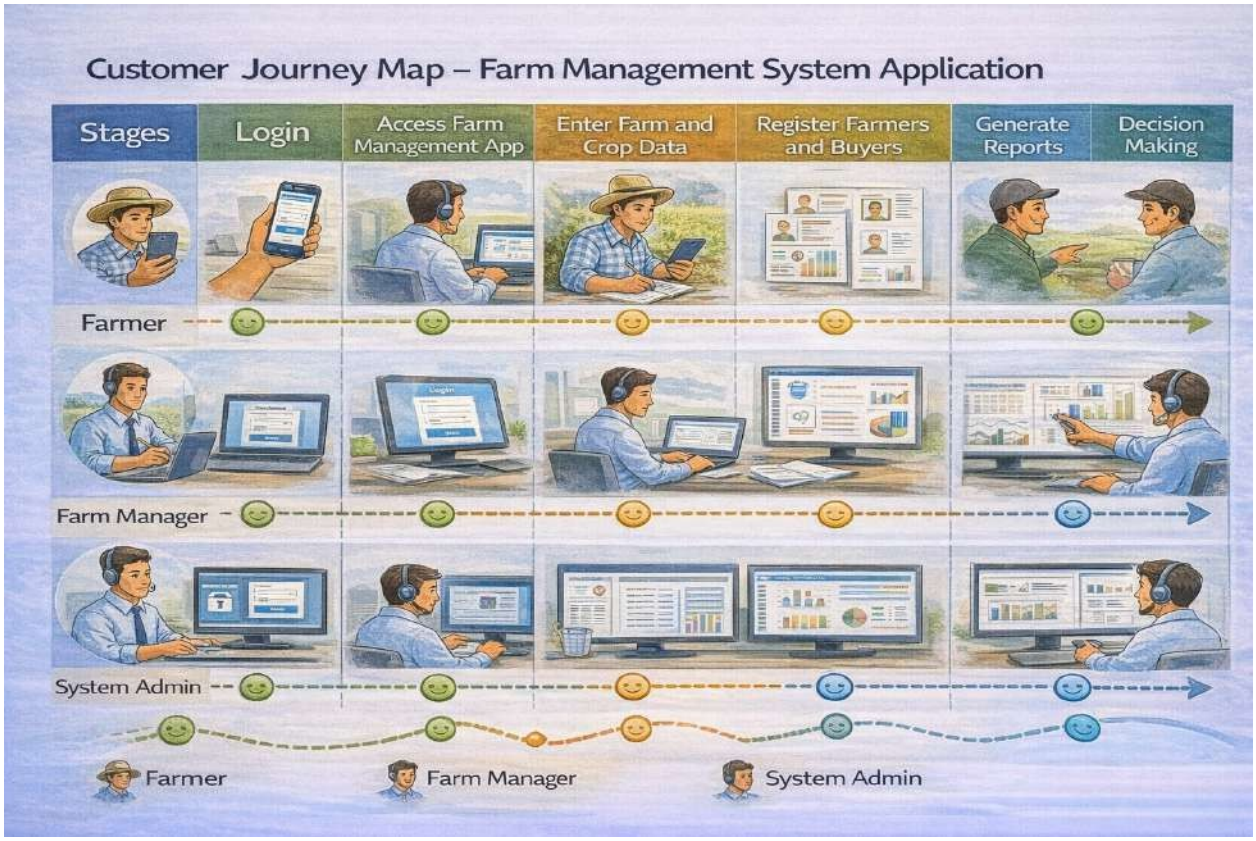


3. REQUIREMENTANALYSISPHASE

CustomerJourneyMap

Stages:

1. Login
2. AccessFarmManagementApp
3. EnterFarmandCrop Data
4. RegisterFarmersandBuyers
5. GenerateReports
6. AnalyzeResults
7. DecisionMaking



SolutionRequirements

FunctionalRequirements

FRNo	Requirement
FR-1	UserAuthentication
FR-2	Farm,Crop,Farmer,BuyerManagement
FR-3	DataEntryandValidation
FR-4	RelationshipMapping
FR-5	ReportGeneration

FR-6	DashboardVisualization
FR-7	UserRoleManagement

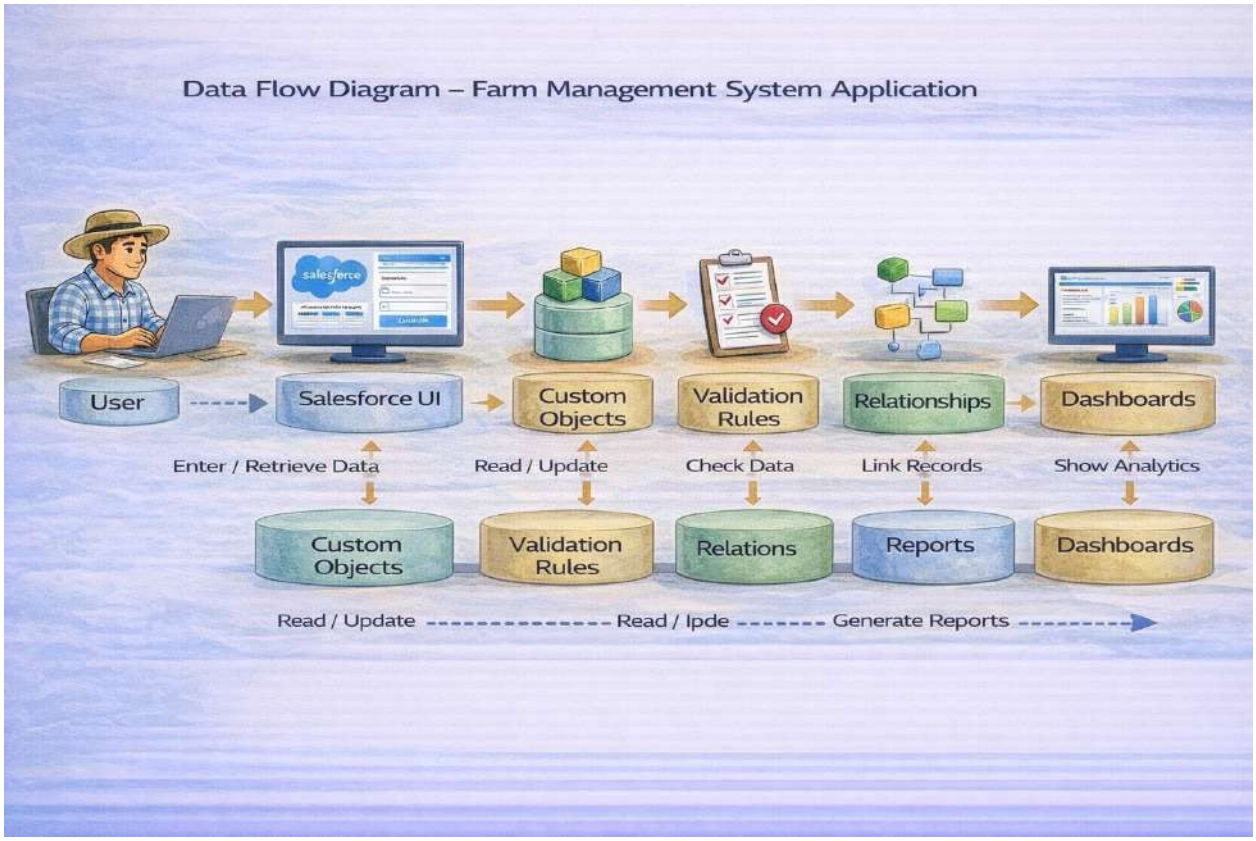
Non-FunctionalRequirements

NFR	Description	Example
Usability	Easyinterface	SimpleLightningUI
Security	Dataprotection	Profiles&roles
Reliability	Nodataloss	Cloudbackup
Performance	Fastaccess	Reportsin<3sec
Availability	24/7access	Cloudhosting
Scalability	Handlesgrowth	Morefarms/users

DataFlowDiagram

DataFlow:

User→SalesforceUI→CustomObjects(Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



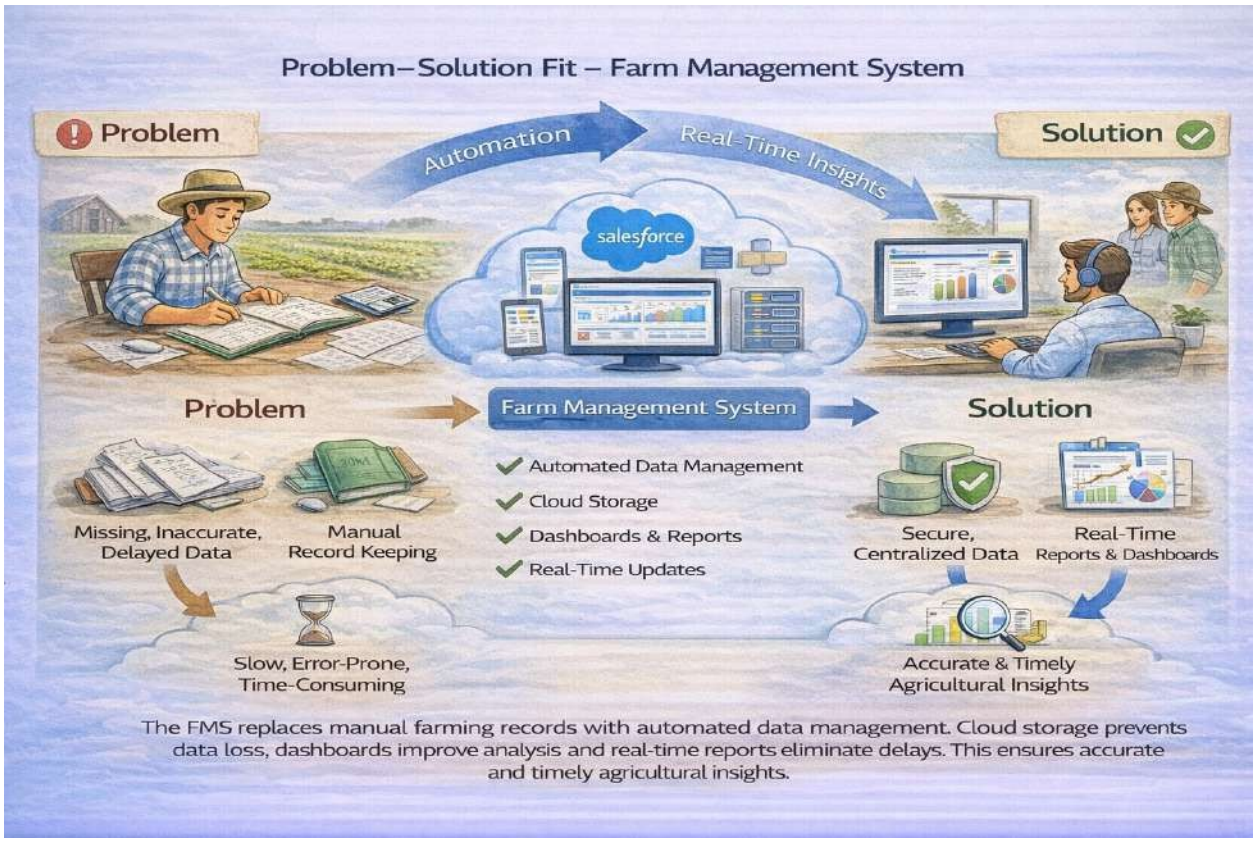
TechnologyStack

Layer	Technology
Frontend	SalesforceLightningUI
Backend	SalesforcePlatform
Database	SalesforceCloud
Automation	ValidationRules
Analytics	Reports&Dashboards

4. PROJECTDESIGNPHASE

Problem–SolutionFit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



ProposedSolution

FunctionalRequirements

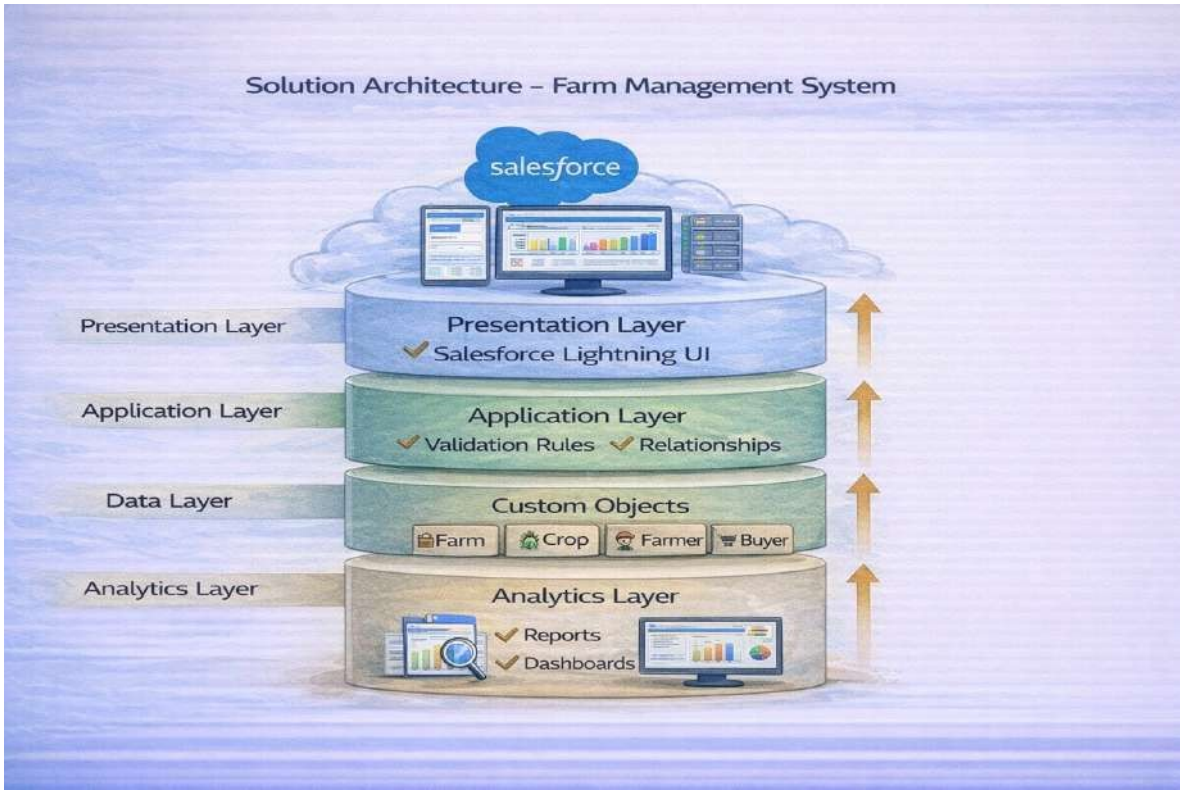
Parameter	Description
Problem	Manualmaintenanceoffarm,crop,farmer,andbuyerrecords leading to errors, data loss, and delays.

Solution	Salesforce-basedFarmManagementSystemwithcentralized data storage and automation.
Innovation	Automateddataentryvalidation,cloudstorage,andreal-time reporting dashboards.
Impact	Improveddataaccuracy,fasterreporting,andbetter decision-making for farmers and managers.es
BusinessModel	Digitalfarmmanagementandagriculturaldataservicesplatform.
Scalability	Supportsmultiplefarms,users,andregionswithcloud-based infrastructure.

SolutionArchitecture

ArchitectureLayers

1. PresentationLayer–SalesforceLightningUI
2. ApplicationLayer–ValidationRules,Relationships
3. DataLayer–CustomObjects
4. AnalyticsLayer–ReportsandDashboards



5. PROJECTPLANNING&SCHEDULING

ProjectPlanning

- CreatedSalesforceDeveloperOrg
- Developedcustomobjects
- ConfiguredtabsandLightningApp
- Createdfieldsandrelationships
- Designedprofilesandroles
- Createdusersandsharingrules
- Developedreportsanddashboards
- Conductedtestingandreviews

ProductBacklog,SprintSchedule,andEstimation(4Marks)

Usethebelowtemplatetocreateproductbacklogandsprintschedule

Sprint	Functional Requirement (Epic)	User Story Number	UserStory/Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	Asasystemadministrator,Iwant to create and configure a Salesforcedeveloperaccountso that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	DataModeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objectswithproperrelationshipsso that agricultural data is stored and managedefficiently.	5	High	Member2
Sprint-2	DataModeling	USN-3	Asastaffmember,Iwanttabsand a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incompleterecordsareprevented.	3	High	Member4
Sprint-3	Automation	USN-5	Asamanager,Iwantautomated flowsforcropupdatesandbuyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-basedaccesscontrolfor farmers,managers,andbuyersso thatsensitivefarmandtransaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	Asafarmmanager,Iwantreports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and makeinformeddecisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operationalstatisticssothatIcan monitor performance visually.	4	Medium	Member4

ProjectTracker,Velocity&BurndownChart:(4Marks)

Sprint	TotalStory Points	Duratio n	SprintStart Date	SprintEndDate (Planned)	StoryPoints Completed (as on PlannedEnd Date)	SprintRelease Date (Actual)
Sprint-1	20	6Days	24Oct2022	29Oct2022	20	29Oct2022
Sprint-2	20	6Days	31Oct2022	05Nov2022		
Sprint-3	20	6Days	07Nov2022	12Nov2022		
Sprint-4	20	6Days	14Nov2022	19Nov2022		

6. ProjectDevelopmentPhase:

ProjectDescription:

TheFarmManagementSystem(FMS)inSalesforceisacomprehensivesolution designedtodigitizeandstreamlineagriculturaloperations.Thisapplicationempowers farmers, farmmanagers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

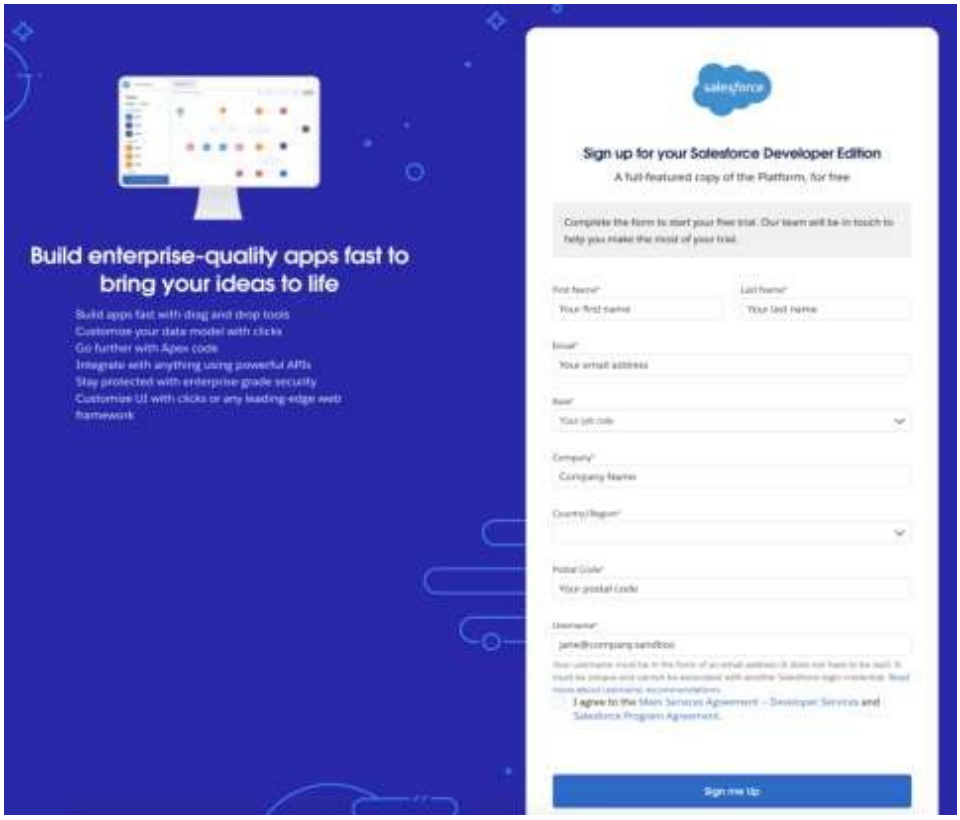
Milestone1-Salesforce:

WhatIsSalesforce?

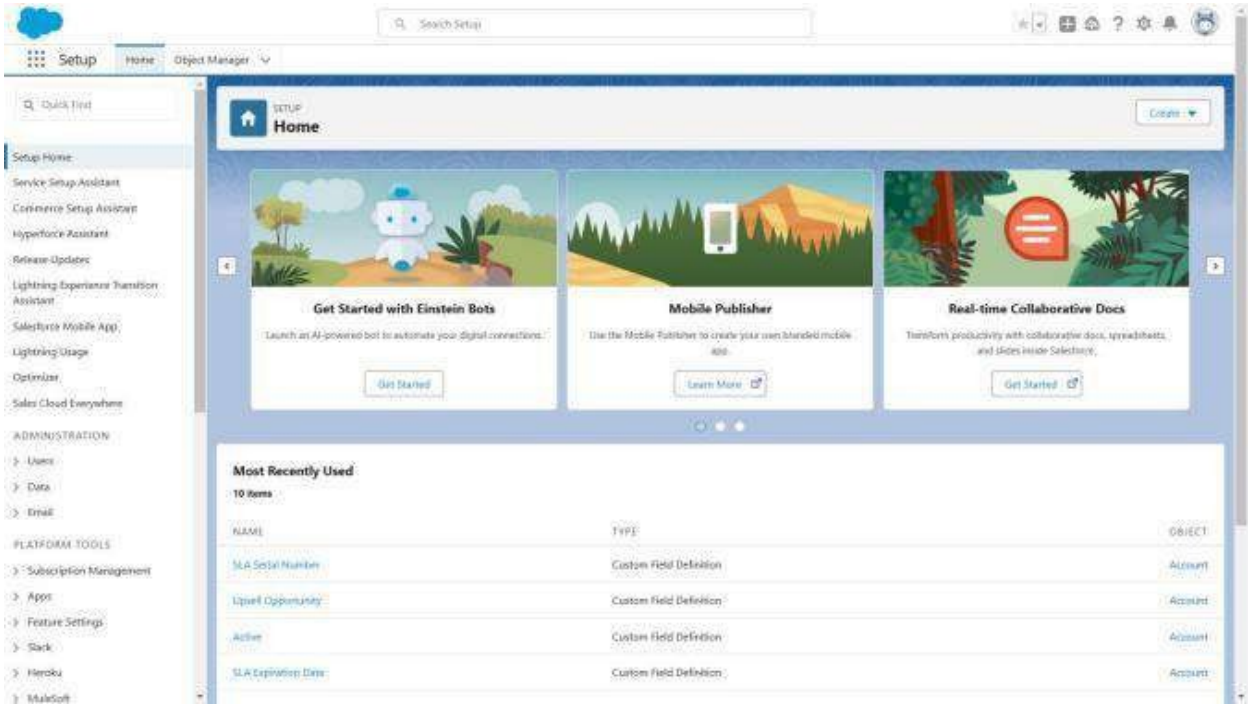
Salesforceisyourcustomersuccessplatform,designedtohelpyousell,service,market, analyze, and connect with your customers. Salesforcehaseverythingyouneedtorunyourbusinessfromanywhere.Usingstandard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:
CreatingaSalesforceDeveloperOrg:

ADeveloperorghasallthefeaturesandlicensesyouneedtogetstartedwithSalesforce.
1. SearchDeveloper.salesforce.com/signup
2. EnterthefollowingdetailslikeFirstname,lastname,Email,Role,Company, Country/Region, Postal code, and Username must be unique.
3. Clicksignmeup,afterafewminyouwillreserveamailsalesforceorgandbyusing the verify account link you can create your new password.
4. Clicksave.
5. Searchlogin.salesforce.com



6. ByusingusernameandpasswordyoucanintotheSalesforceorg.Thehomepagewill appear below



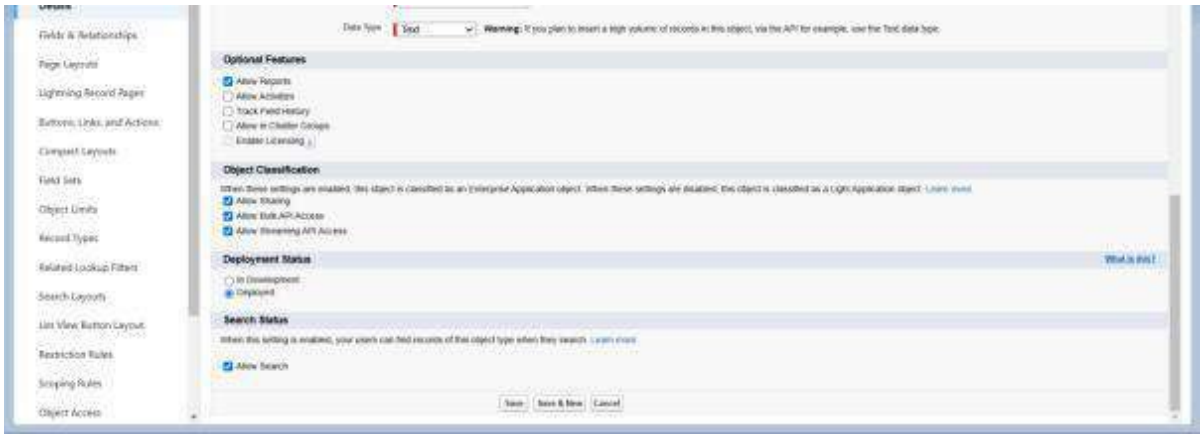
Milestone2-Object:

Whatisan object?
Salesforceobjectsaredatabasetablesthatpermityoutostoredatathatisspecifictoan organization. It consists of fields (columns) and records (rows).

InThisApplicationWeUse4CustomObjects:
1.Farm
2.Crop
3.Farmer
4.Buyer

Activity1:
CreateacustomobjectforFarm:
Tocreateacustomobject,followthesesteps:
1. Fromsetupclickonobjectmanager.
2. Clickcreate,selectcustomobject.

3. Fillinthelabelas"Farm".
4. Fillintheplurallabelas"Farms".
5. Recordname:FarmName"
6. Selectthedatatypeas"Text".



7. IntheOptionalFeaturessection,selectAllowReportsandTrackFieldHistory.
8. IntheDeploymentStatussection,ensureDeployedisselected.
9. IntheSearchStatussection,selectAllowSearch.
10. IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout.

11. Leaveeverythingelseasis,andclickSave.

Activity2:

CreationofCropobject:
1. ClickonthegeariconandthenselectSetup.
2. Clickontheobjectmanagertabjustbesidethehome tab.
3. Aftertheabovesteps,havealookontheextremerightyouwillfindaCreate Dropdown click on that and select Custom Object.
a. OntheCustomObjectDefinitionpage,createtheobjectasfollows:
b. Label:Crop
c. PluralLabel:Crops
d. RecordName:CropName

e. Selectthedatatypeas"Text".
f. CheckktheAllowReportscheckboxg.CheckktheAllowSearchcheckbox
h.IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout i. Click Save.

Activity3:

CreationofFarmerobject:
1. ClickonthegeariconandthenselectSetup.
2. Clickontheobjectmanagertabjustbesidethehome tab.
3. Aftertheabovesteps,havealookontheextremerightyouwillfindaCreate Dropdown click on that and select Custom Object.
a. OntheCustomObjectDefinitionpage,createtheobjectasfollows:
b. Label:Farmer

c. PluralLabel:Farmers
d. RecordName:FarmerName e.Selectthedatatypeas"Text".
f.Checkthe AllowReportscheckbox.g.Checkthe AllowSearchcheckbox.
h.IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout. i. Click Save.

Activity4:

CreationofBuyerobject:
1. Clickonthegeariconandthenselect Setup.
2. Clickontheobjectmanagertabjustbesidethehome tab.
3. Aftertheabovesteps,havealookontheextremerightyouwillfindaCreate Dropdown click on that and select Custom Object.
a. OntheCustomObjectDefinitionpage,createtheobjectasfollows:
b. Label:Buyer
c. PluralLabel:Buyers
d. RecordName:BuyerName
e. Selectthedatatypeas"Text".
f. Checkthe AllowReportscheckbox.g.Checkthe AllowSearchcheckbox.
h. IntheObjectCreationOptionssection,selectAddNotesandAttachmentsrelatedlist to default page layout.
i. ClickSave.

Milestone3- Tab:
Tab

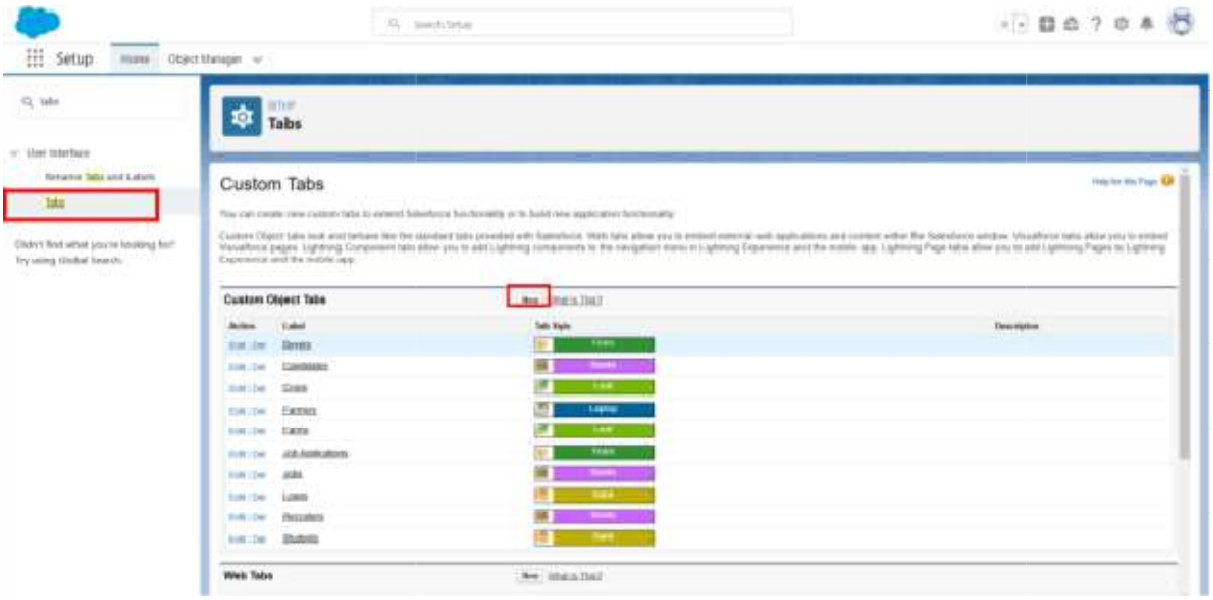
TabsinSalesforcehelpusersviewtheinformationataglance.Itdisplaysthedataof objects and other web content in the application.

Activity1:

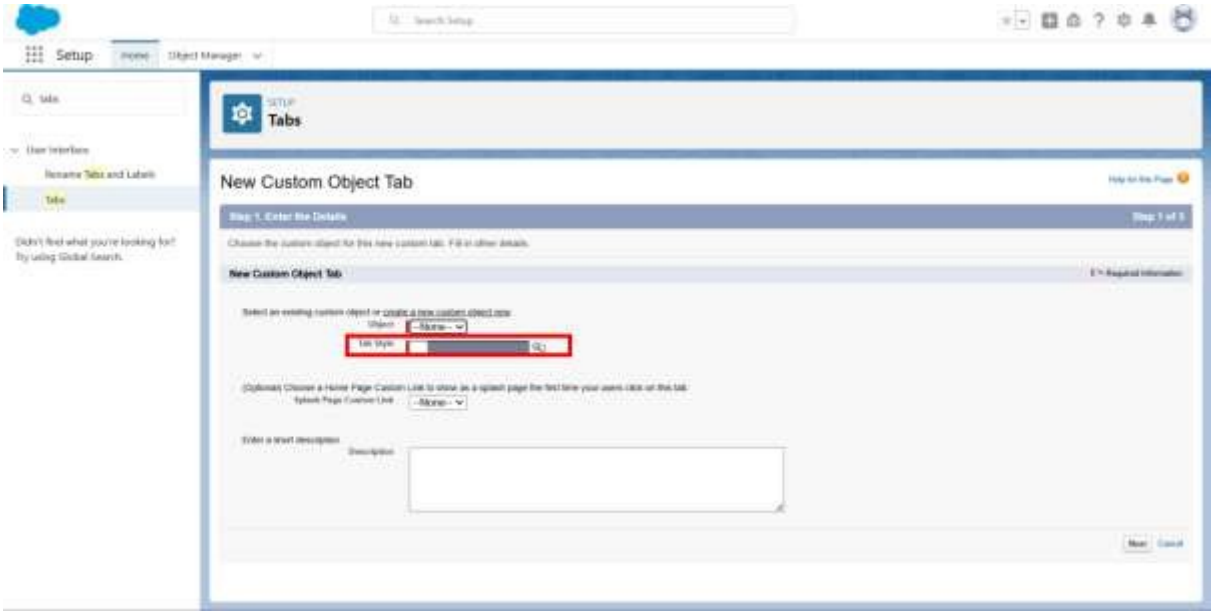
CreationofFarmtab:

Nowcreateacustom tab.

1. ClickontheHometab,enterTabsinQuickFindandselectTabs.



2. Undercustomobjecttabs,clickNew.



3. ForObject,selectFarm.

4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

Activity2:

CreationofCroptab:
Nowcreateacustom tab.
1. ClickonHometab,enterTabsinQuickFindandselectTabs.
2. Undercustomobjecttabs,clickNew.
3. ForObject,selectCrop.

4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

Activity3:

CreationofFarmertab:

Nowcreateacustom tab.
1. ClickontheHometab,enterTabsinQuickFindandselectTabs.
2. Undercustomobjecttabs,clickNew.
3. ForObject,selectFarmer.
4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

Activity4:

CreationofBuyertab:

Nowcreateacustom tab.
1. ClickonHometab,enterTabsinQuickFindandselectTabs.
2. Undercustomobjecttabs,clickNew.
3. ForObject,selectBuyer
4. ForTabStyle,selectanyicon.
5. Leavealldefaultsasis.ClickNext,Next,andSave.

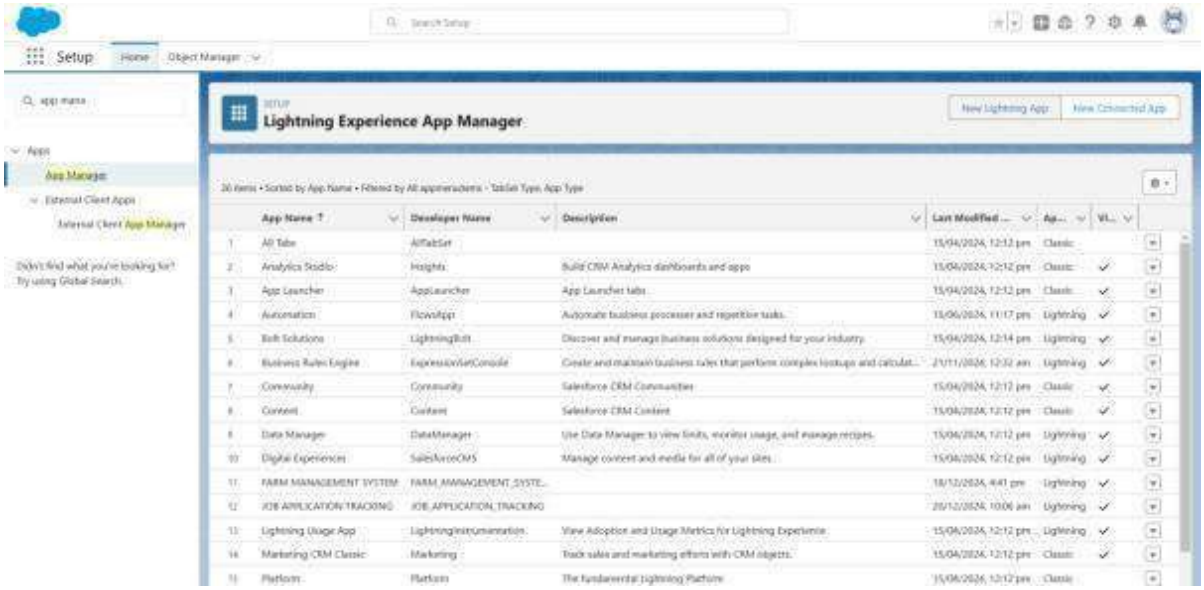
Milestone4-LightningApp:

WhatisanApp?

Apps in Salesforce area group of tabs that help the application function by working togetherasaunit.Ithasaname,alogo,andaparticularsetoftabs.Thesimplestapp usually has justtwo tabs.

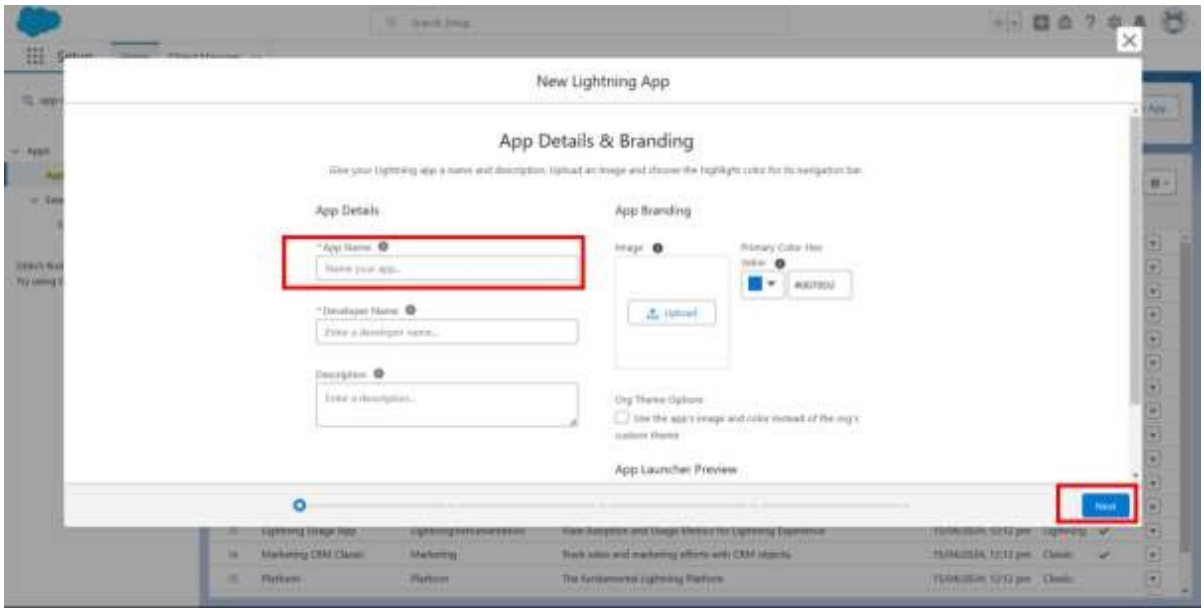
Activity1:

CreatetheFarmManagementSystemapp

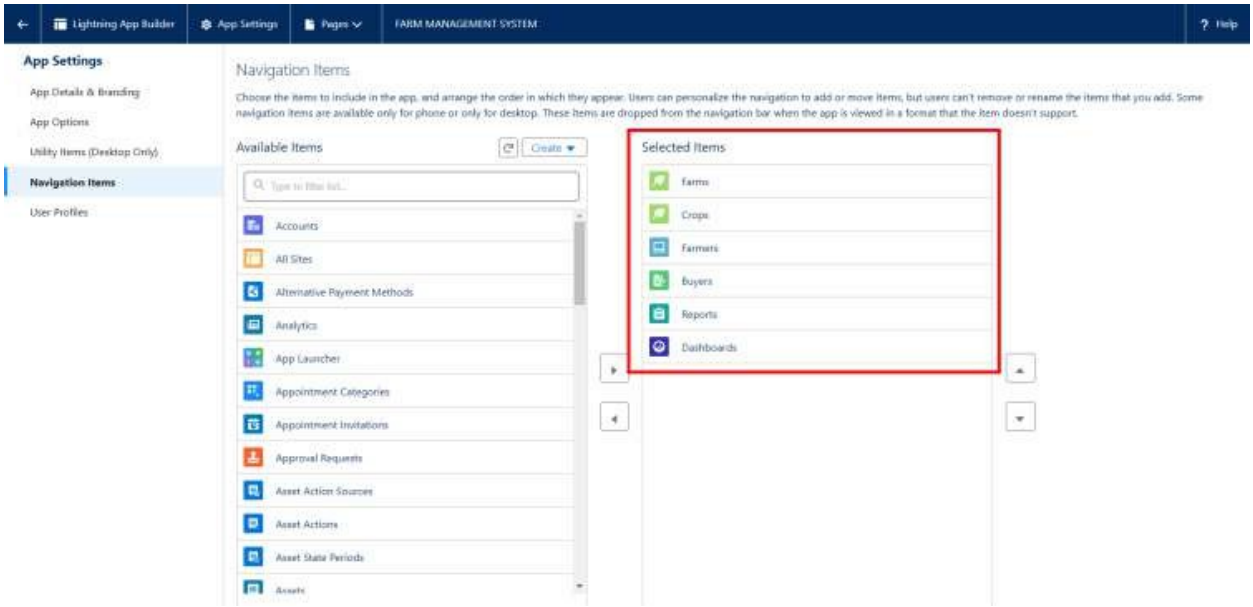


● FromSetup,enterAppManagerintheQuickFindandselectAppManager.

● ClickNewLightningApp.FarmManagementSystemastheAppName,thenclick Next
● UnderAppOptions,leavethedefaultselectionsandclickNext.
● UnderUtilityItems,leaveasisandclickNext.
● FromAvailableItems,selectFarm,Crop,Farmer,Buyer Reports,andDashboardsandmovethemtoSelectedItems.ClickNext.●FromAvailable Profiles, select System Administrator and move it toSelected Profiles.Click Save &Finish.
● Toverifyyourchanges,clicktheAppLauncher,typeFarmManagementSystemand select the Job Application app.
Note:
1. AppLauncher-Displaysavailableapps.
2. AppName-Displaysthecurrentselectedapp.



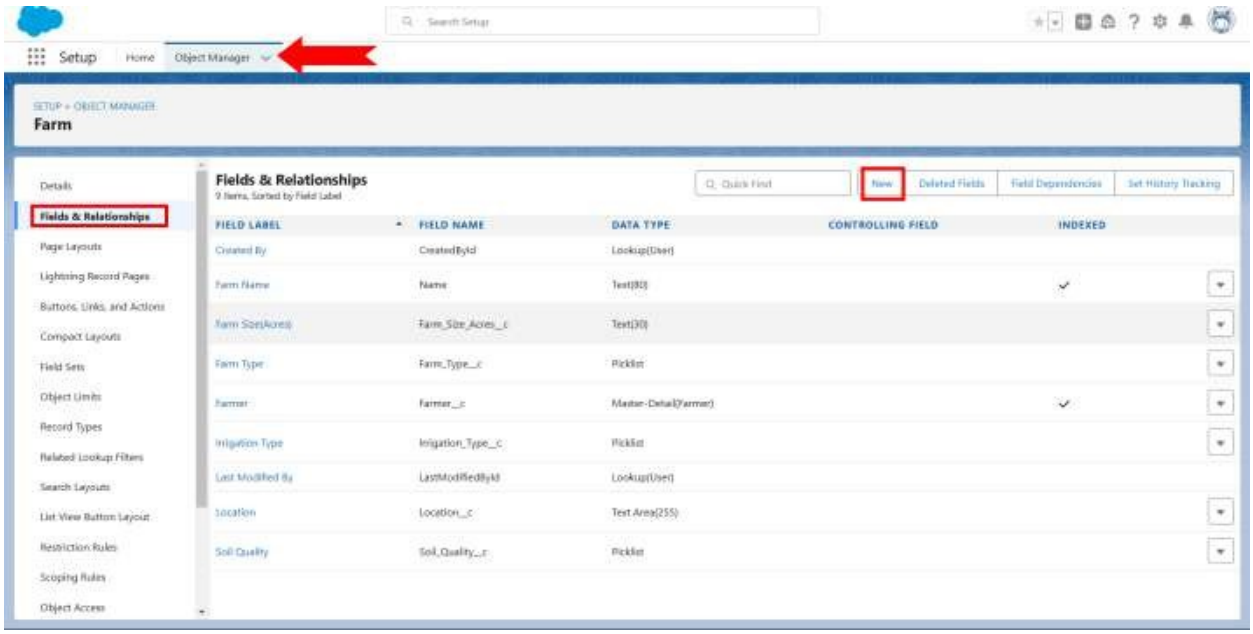
3. Navigationmenu-Displaysthetabsavailableinsidetheapp.



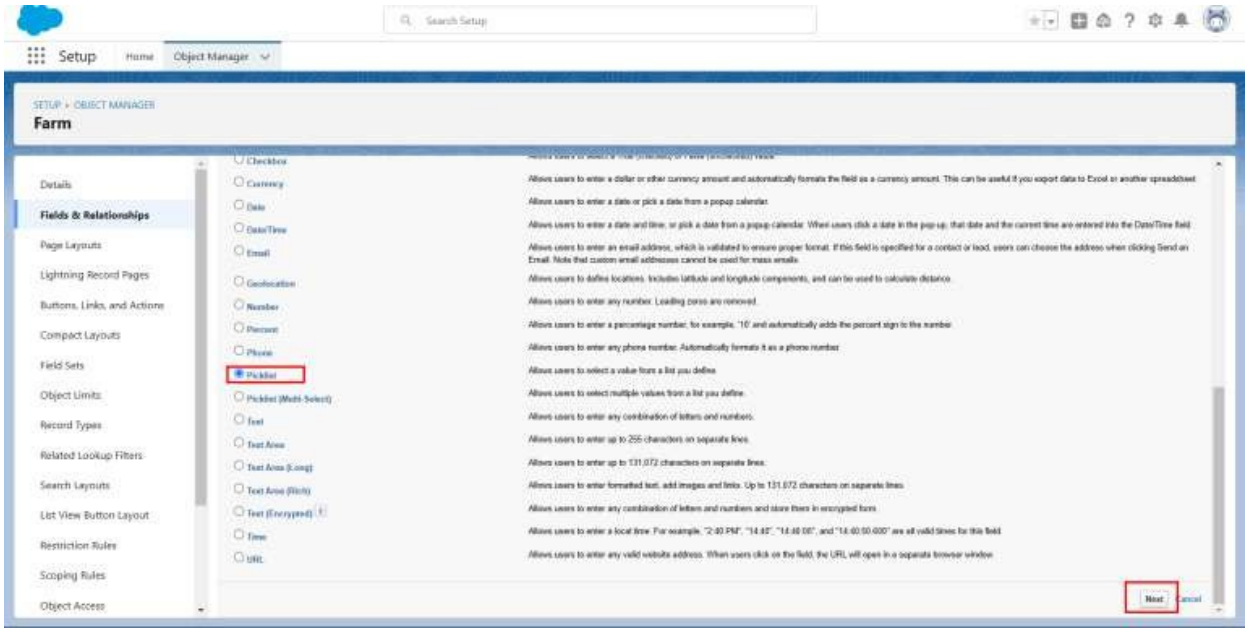
Milestone5-
FieldsandRelationship:Whatarefields?

FieldsinSalesforcerepresentwhatthecolumnsrepresentinrelationaldatabases.Itcan store data values which are required for a particular object in a record.

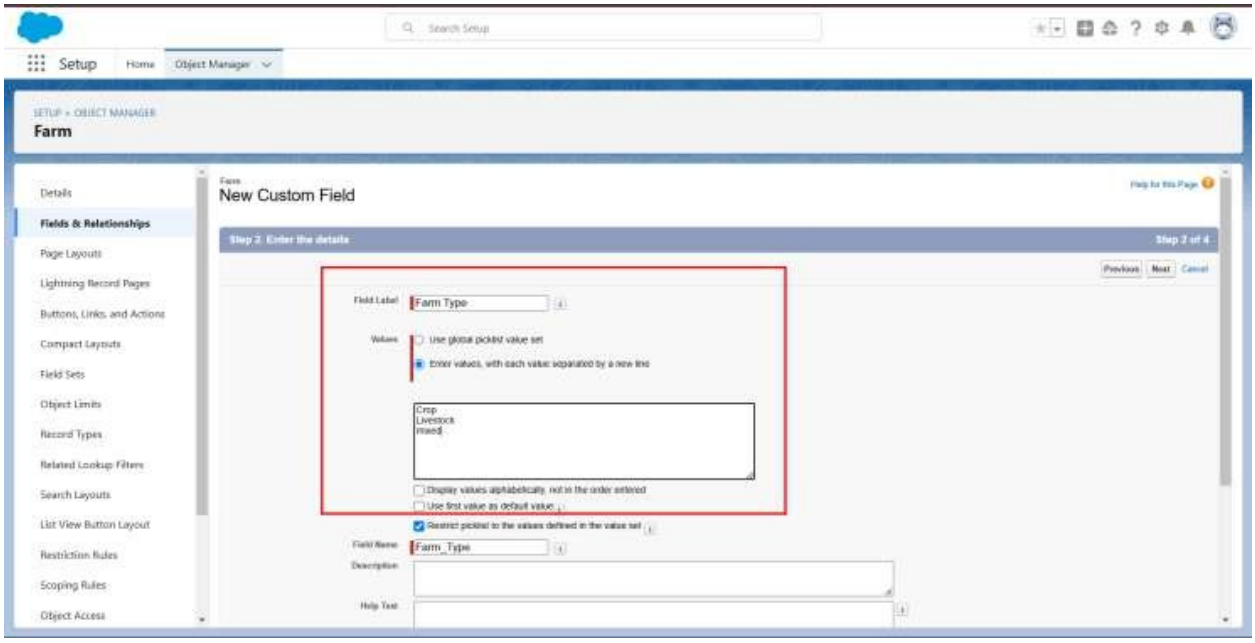
Activity1:
CreationoffieldsfortheFarmobjects:
1. clickthegeariconandselectSetup.ThislaunchesSetupinanew tab.
2. ClicktheObjectManagertabnexttoHome.
3. SelectFarm.
4. SelectFields&Relationshipsfromtheleftnavigation,andclick New



5. ChooseDatatypeasapicklist,Clicknext



6. EnterFieldLabel(FarmType),andEnterValuesas(Crop,Livestock,Mixed)
7.Next,Next Click Save.

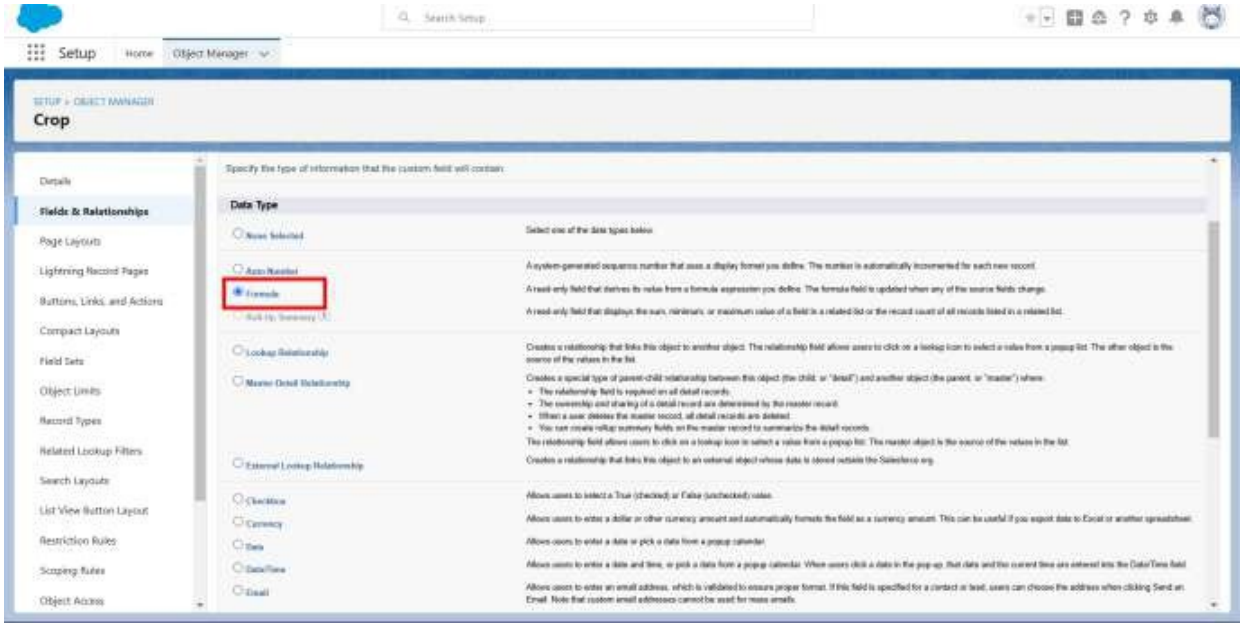


Nowlet'screatetheotherfieldsfollowingActivity1steps1to4andwemustchoosethe data types of the fields carefully.

1. NextweCreateanotherPicklistField,Enterfieldlabel(SoilQuality)entervalues (Organic,Cover Crops,Composting)
2. AgainweCreatePicklistfield,FieldLabel(Irrigationtype),EnterValues(Sprinkler irrigation,Drip irrigation,localized irrigation)
3. Choosedatatypeasatext,enterfieldlabel(Location)4.Choosedatatypeastext,enter field label (size)
5. CreateaMasterDetailrelationShipwithFarmerObject.

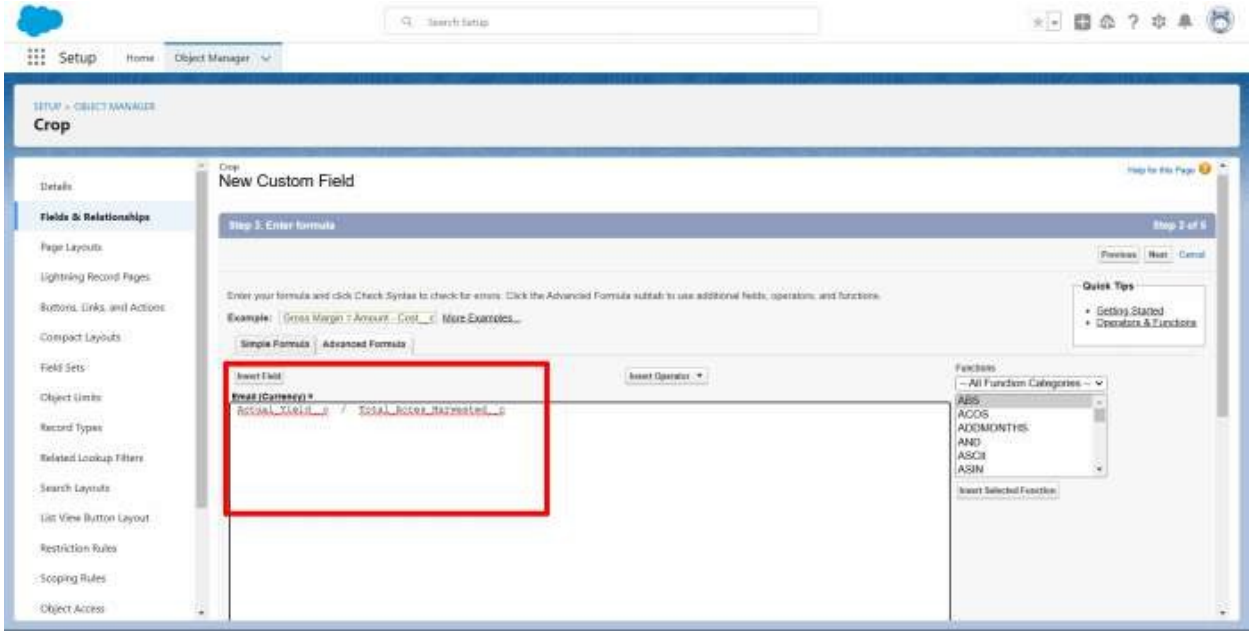
Activity2:
CreationoffieldsfortheCropobjects:

1.ChooseDatatypeAsatext,EnterFieldLabelname(CropName),Nextsave.2.Select Data type as date ,Enter Field label (Planting Date),then save .
3. Againweselectdatatypeasadate,Enterfieldlabe(Harvestdate),save
4. NextWeCreateLookuprelationshiponcropobjectrealtedtoFarmObject.
5. Weselectadatatypeascurrency,EnterfieldLabel(ActualYield),NextSaveit.
6. SelectDatatypeasaNumber,enterFieldLabel(TotalAcresHarvested),Enter Maximum Length(20) and save.
7. NowWecreateFormulaFieldForThatSelectdataTypeAsaFormula,ClickNext.


Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Crop
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Specify the type of information that the custom field will contain
Data Type
None Selected
Select one of the data types below.
Auto Number
A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula
A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary
A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship
Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship
Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:
• The relationship field is required on all detail records.
• The ownership and sharing of a detail record are determined by the master record.
• When a user deletes the master record, all detail records are deleted.
• You can create rollup summary fields on the master record to summarize the detail records.
The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
External Lookup Relationship
Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.
Checkbox
Allows users to select a True (checked) or False (unchecked) value.
Currency
Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
Date
Allows users to enter a date or pick a date from a popup calendar.
Date/Time
Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.
Email
Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.

8. EnterFieldLabel(ExpectedYieldperAcre),Selectformulareturntypeas Currency,click Next.

9. EnteraFormulaActualYieldDivideByTotalAcreHarvested,thenclicknextand save.



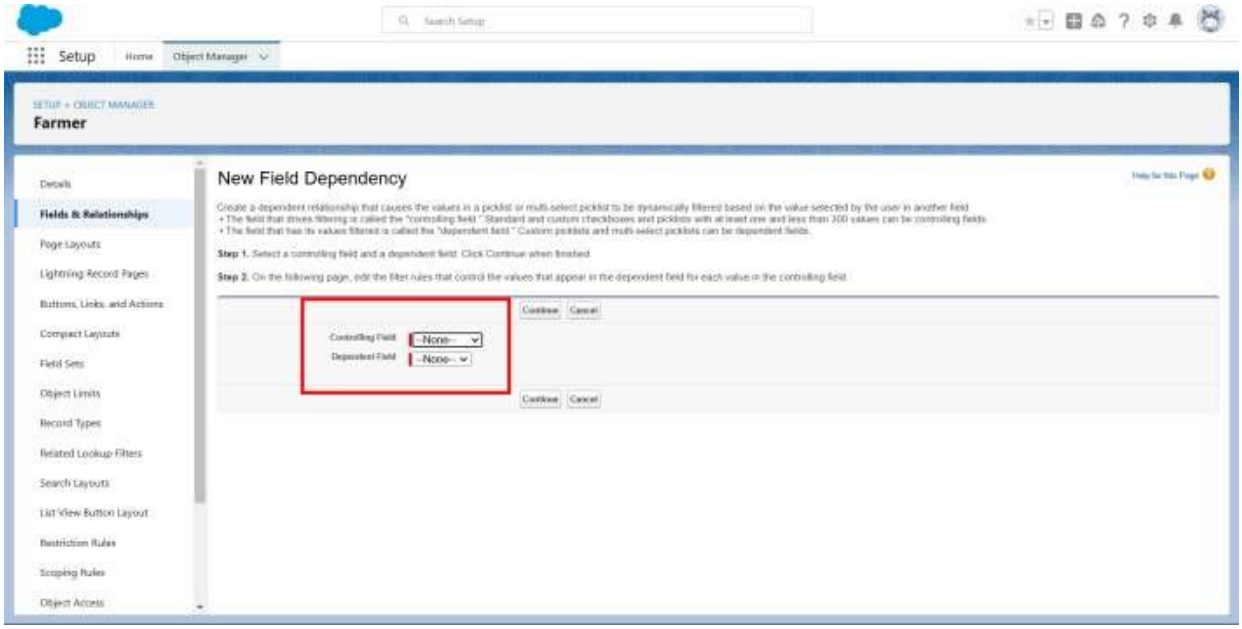
Activity3:
CreationoffieldsfortheFarmerobjects:

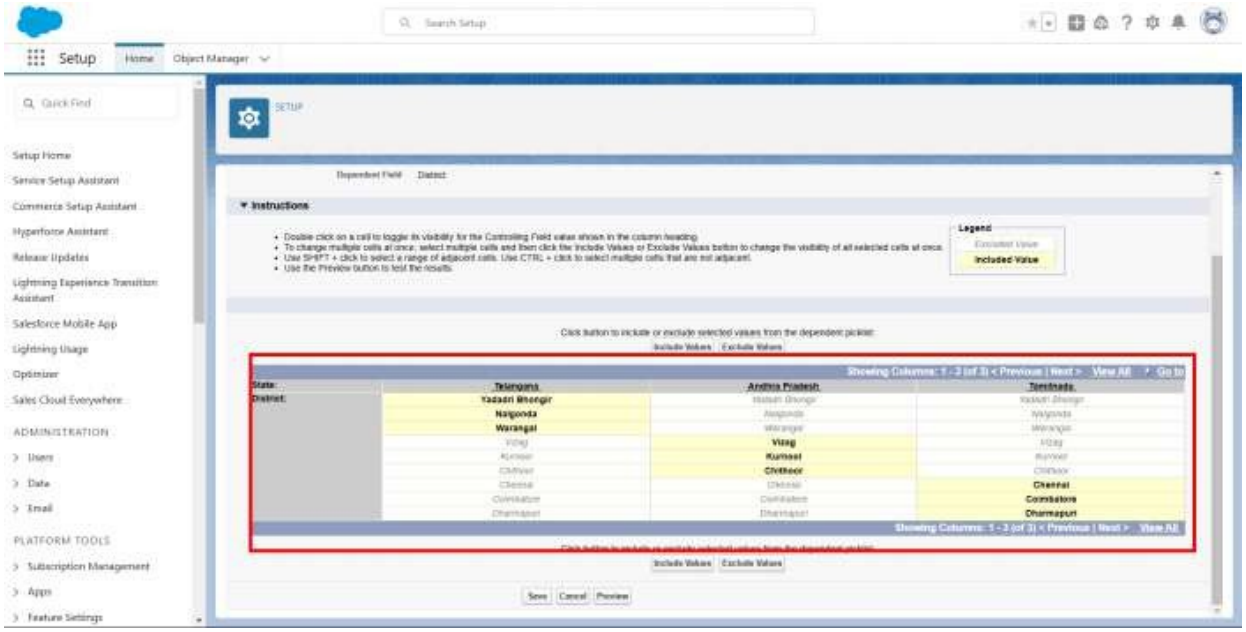
1. SelectDatatypePhone,Enterfieldlabel(ContactNumber),nextSave.
2. SelectDatatypePicklist,EnterFieldLabel(State),Entervalues(Telangana,Andhra Pradesh,Tamilnadu),save.
3. ChooseDatatypePicklist,EnterFieldLabel(District),EnterValuesofdistrict,nextsave.
4. AgainselectDatatypeasPicklist,EnterFieldLabel(Village),EnterValuesof villages,next save.

Creationoffielddependency:

1. NowcreateFieldDependencyForDistrictfield,OpenObjectManager,ChooseFarmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.
2. Clicknew,SelectControllingFieldasState,DependentFieldAsDistrict,Clickon Continue.
3. Selecttheappropriate Valueineachcolumnbydouble-clickingthem.
4. ForEx.Telangana-YadadriBhongir.
5. ClickIncludeValues.AnditisalsothesameforAndhraPradesh,Tamilnaduwithits city.

6. ClickPreview,thentestthedependencybyselectingadifferentStateandviewingthe associate Values available for Particular state.


Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Farmer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
New Field Dependency
Controlling Field
--None--
Dependent Field
--None--
Continue
Cancel


Search Setup
Setup
Home
Object Manager
Quick Find
Setup Home
Service Setup Assistant
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Lightning Experience Transition Assistant
Salesforce Mobile App
Lightning Usage
Optimizer
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
PLATFORM TOOLS
Subscription Management
Apps
Feature Settings
SETUP
Dependent Field District
Instructions
Legend
Included Value
Click button to include or exclude selected values from the dependent picklist:
Include Values
Exclude Values
State:
District:
Telangana
Andhra Pradesh
Tamilnadu
Yadadri Bhongir
Nalgonda
Warangal
Vizag
Kurnool
Chittoor
Chennai
Coimbatore
Dharmapuri
Save
Cancel
Preview

7. ClickClosetoclosethepreviewwindow,ClickSave.

Activity4:

CreationoffieldsfortheBuyerobjects:

1. SelectDatatypeasaPhone,EnteraFieldLabel(Contact),Save.
2. SelectDatatypeasanEmail,EnterFieldLabel(Email),Nextsave.
3. SelectDatatypeTextarea,EnterFieldLabel(Address),Saveit.
4. SelectDatatypePicklist,FieldLabel(BuyerType),Entervalues (Wholesale,Retail),Click Save.

Milestone6-Profile:

Whatisa profile?

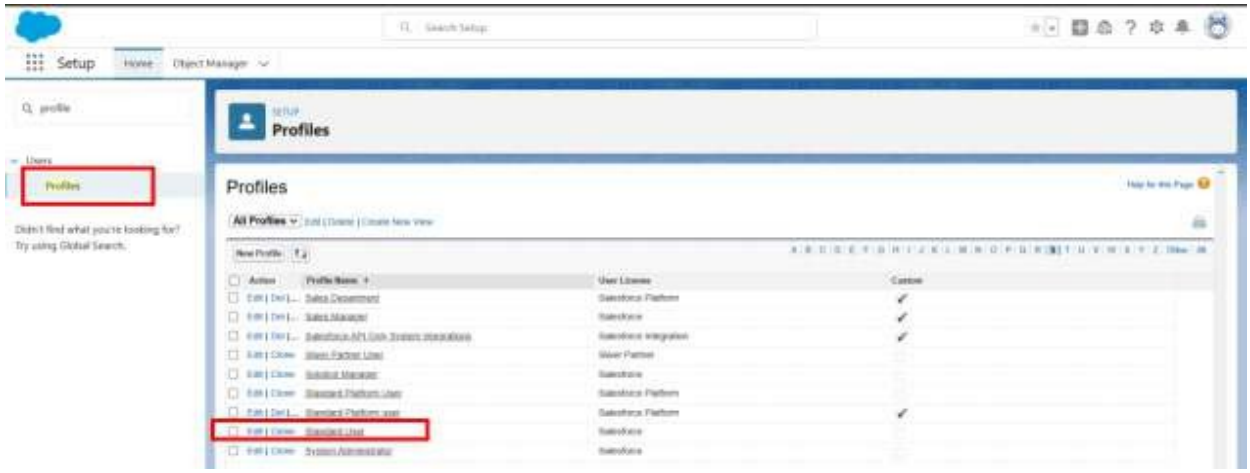
Aprofileisagroup/collectionofsettingsandpermissionsthatdefinewhatausercando in

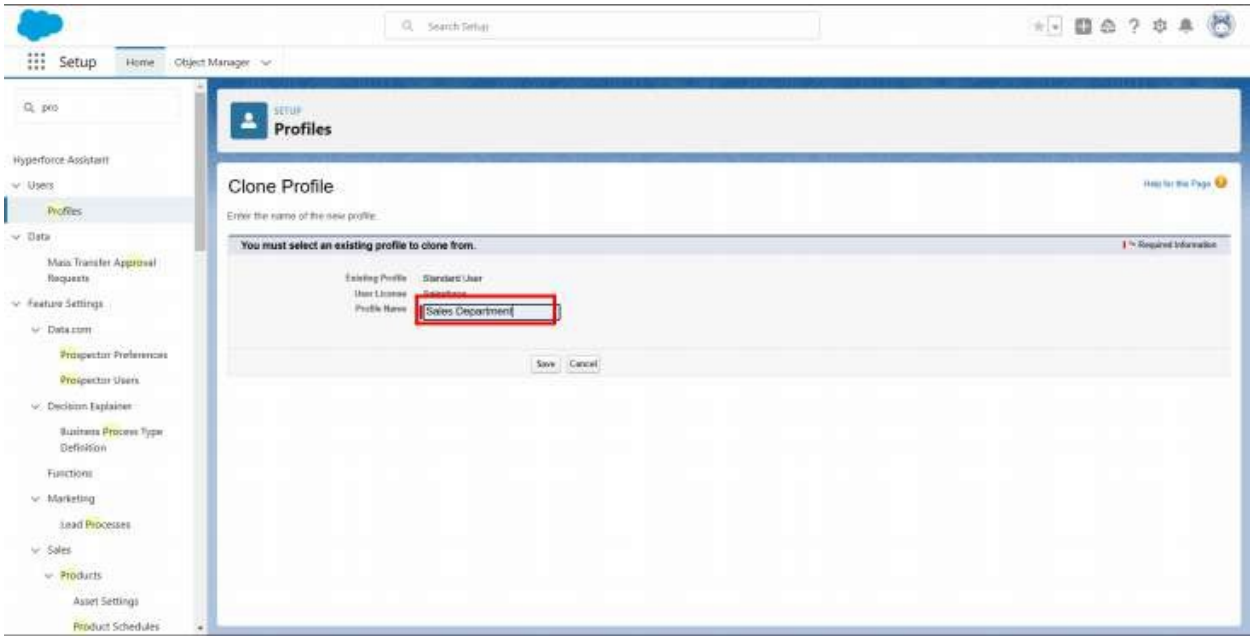
salesforce.Aprofilecontrols“Objectpermissions,Fieldpermissions,Userpermissions,
Tab
settings,Appsettings,Apexclassaccess,Visualforcepageaccess,Pagelayouts,Record Types,
Login hours & Login IP ranges.
Aprofilecanbeassignedtomanyusers,butusercanbeassignedsingleprofileatatime.

Activity1:

Createacustomprofile:

1. Fromsetup,enterprofilesinQuickFindbox
2. Selectprofiles(Standarduser).
3. Clickclone.
4. ForProfile,enterSalesDepartment.


Setup
Home
Object Manager
profile
Users
Profiles
Didn't find what you're looking for?
Try using Global Search.
SETUP
Profiles
Profiles
All Profiles
Action
Profile Name
User License
Custom


Setup
Home
Object Manager
Search Setup
pro
Hyperforce Assistant
Users
Profiles
Data
Mass Transfer Approval Requests
Feature Settings
Data.com
Prospector Preferences
Prospector Users
Decision Explainer
Business Process Type Definition
Functions
Marketing
Lead Processes
Sales
Products
Asset Settings
Product Schedules
SETUP
Profiles
Clone Profile
Help for this Page
Enter the name of the new profile.
You must select an existing profile to clone from.
Required Information
Existing Profile
Standard User
User License
Salesforce
Profile Name
Sales Department
Save
Cancel

5. Clicksave.

Activity2:

Createacustomprofile-2:
Createaprofilewiththeprofilenameas“FarmManager”.

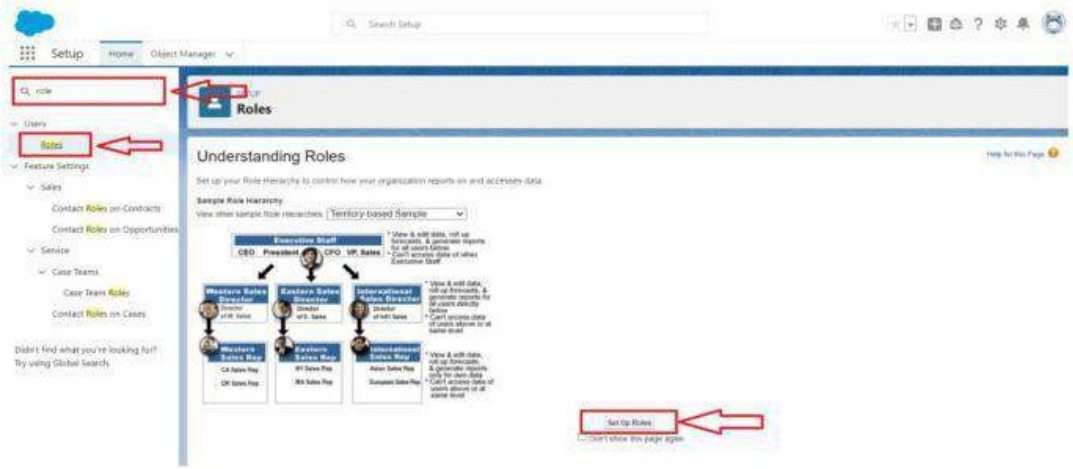
6. Fromsetup,enterprofilesinQuickFindbox
7. Selectprofiles(Standarduser).
8. Clickclone.

Milestone7-Role

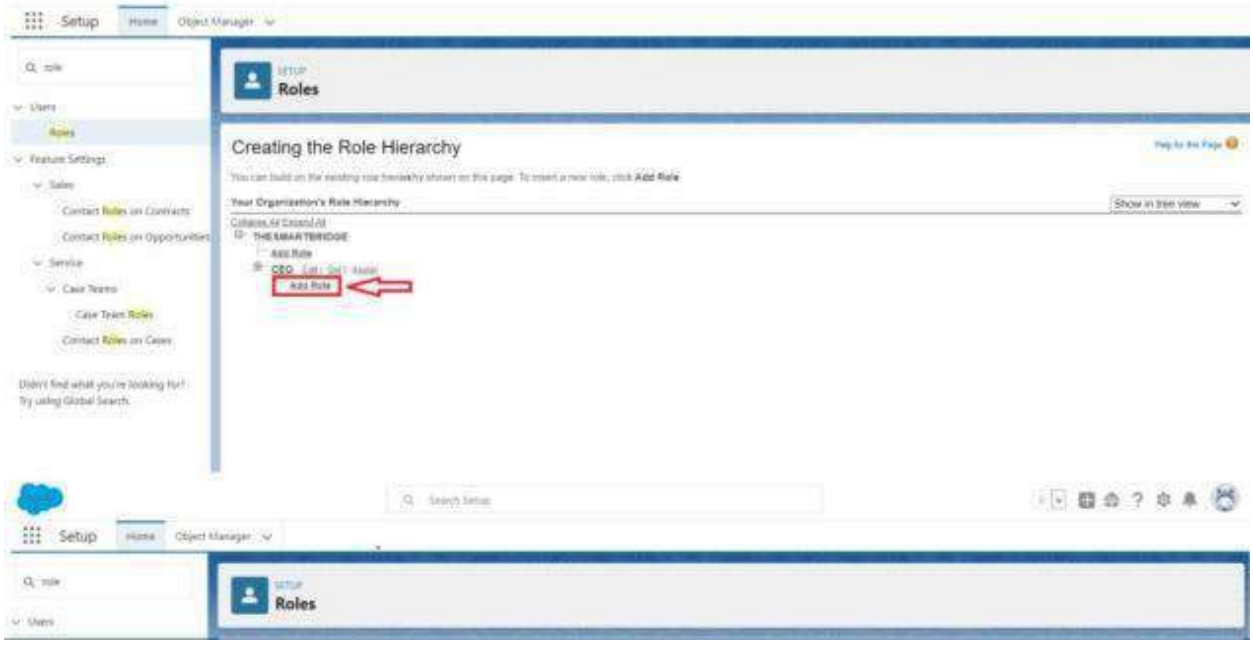
In Salesforce, roles are used to determine which users have access to certain data and functionswithinthesystem.Theyarealsousedtodefinethereportinghierarchywithin an organization.Users with higher roles have greater access to data and more control over the system.

Activity-1
Creationofrole

1. FromtheQuickfindboxsearchfortheroleandclickontherolesoption
2. selectthesetuprolesoption



3. BelowtheCEOclickonaddroleandenterthelabelnameasa”FarmManager”and role name will be Automatically populated and click on save.



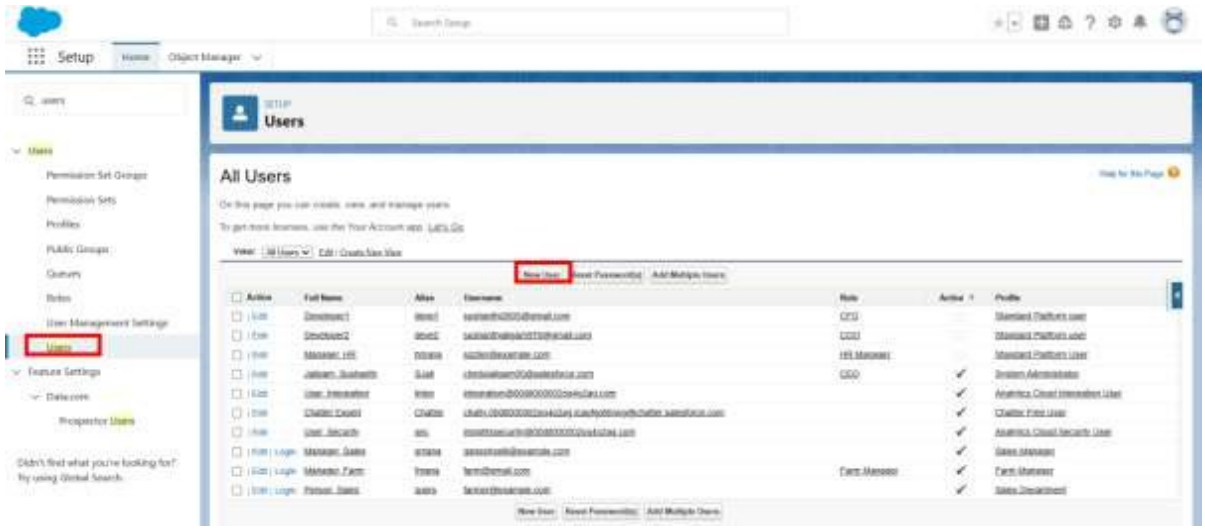
Milestone8-User:

Whatisa user?
AuserisanyonewhologsintoSalesforce.Usersareemployeesatyourcompany,suchas salesreps,managers,andITspecialists,whoneedaccesstothecompany'srecords.Every user in Salesforcehasauseraccount.Theuseraccountidentifiestheuser,andtheuseraccount settings determine what features and records the user can access.

NOTE-AsSalesforcelicensecanonlybeusedby2UsersatatimeinDevOrg,soIfyou don't find salesforcelicensethendeactivateauserwhohassalesforcelicenseOrchangethelicense type from Salesforce to any other.

Activity1:

ToCreateauser:
1. FromSetup,enterUsersintheQuickFindbox,thenselectUsers.
2. ClickNew User.



3. EnterFirstnameasFarmandlastnameasManager.
4. Entertheuser'snameandemailaddressandauniqueusernameintheformofanemail address. By default, the username is the same as the email address.
5. SelecttheroleFarmManager.
6. SelectuserLicenseasStandardPlatformUser.
7. Selectprofile(FarmManager).
8. Clicksave.

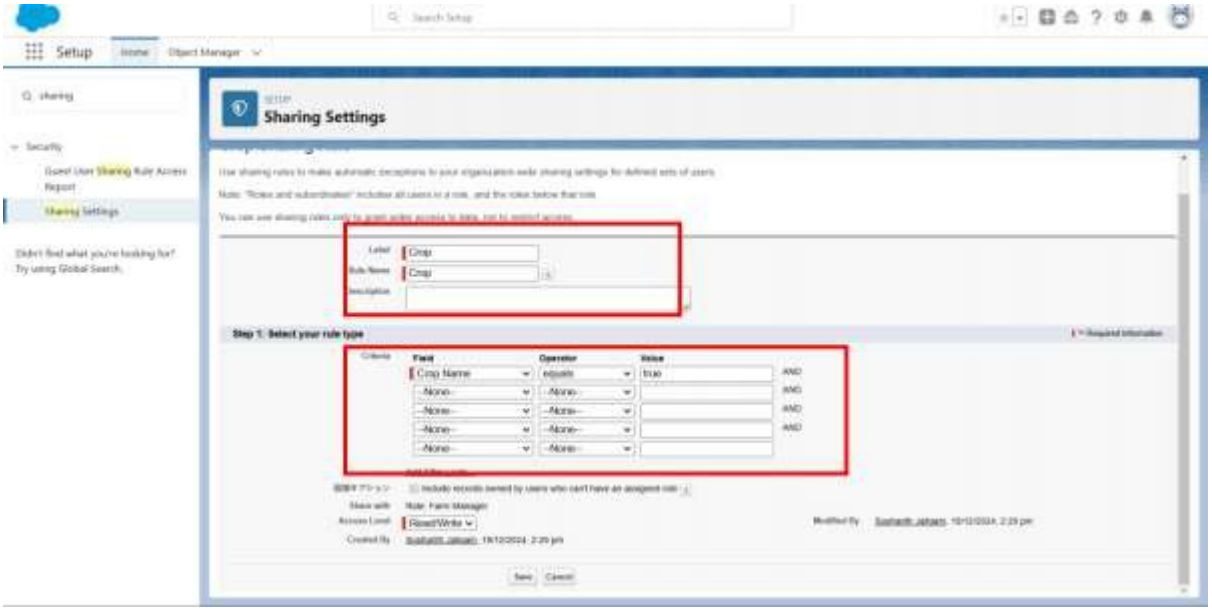
Activity2:
Createauserwithausernameas"SalesDepartment",andassignhimthesalesManager profile.
Follow the steps from above Activity

Milestone9-SharingRules:

WhatareSharingRules?
Sharing rules help users to share records based on conditions. It is basically created for objectswhoseorganization-widedefaults(OWD)aresettopublicread-onlyorprivate because sharing rules can only extend the access and not restrict it.
Typesofsharingrules,
•Owner-basedSharingRules•Criteria-basedSharingRules

Activity1:

Createasharingrule
1) GotoSharingSettings,whichcanbefoundundertheQuickFindsection.
2) Scrolldownandfindthecandidateobjectwhereasharingruleneedstobeadded,and then click on New to create a new sharing rule.
3) Addthelabelofthesharingruleyouwanttomake.
4) Selectyourruletypebasedonthecriteria.
5) Selectthefieldcanjoinimmediatelycheckfieldfromthe Cropobject.
6) SelecttheCropasequalandvalueisTrue.
7) Andinselectingtheuserstosharewiththesectionselectrolesandinthatselect Farm Manager.
8) Andinthesectionofselectthelevelofaccessfortheusersgivetheaccess Read/Write.
9) Andsavetherule.


Setup
Home
Object Manager
Sharing Settings
Security
Sharing Settings
Crop
Step 1: Select your rule type
Field
Operator
Value
Crop Name
equals
true
AND
Read/Write
Save
Cancel

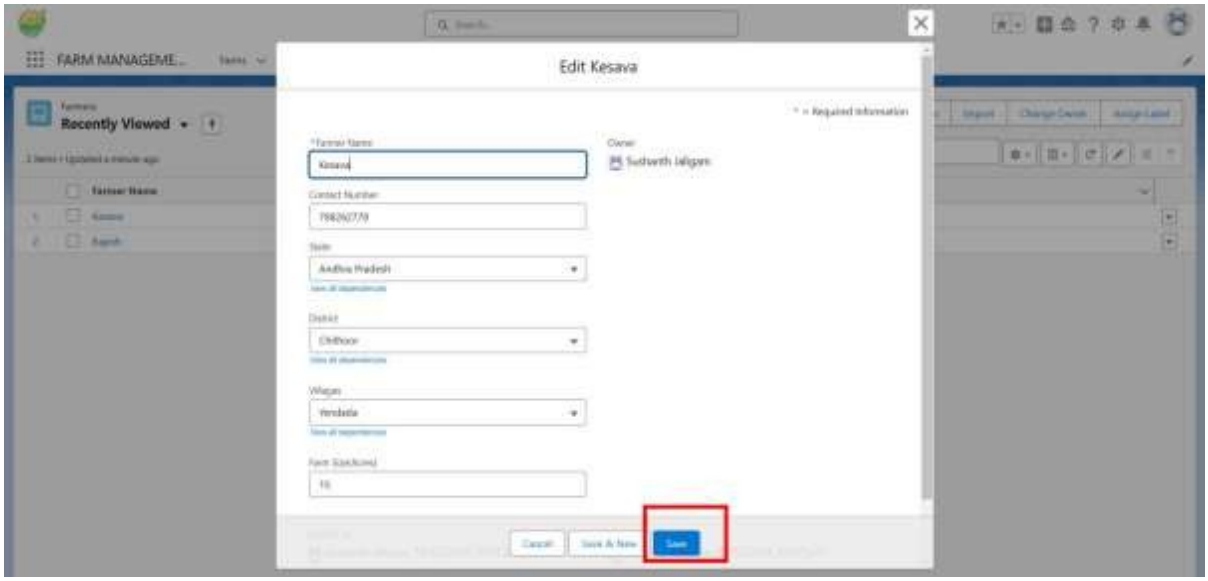
Activity2:

CreateaSharingRuletoSharetherecordsofBuyertoSalesdepartmentwiththeAccess of Read/Write.
FollowthestepsfromaboveActivity.

Milestone-10: User Adoption

Activity1:CreateaRecord(Farmer)

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagement&clickonit.
- ClickonFarmer Tab.
- Clicknewandfilldetails&Save.



Milestone 11-Reports:

WhatareReports?

A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is storedinafolder.Folderscanbepublic,hidden,orshared,andcanbesettoread-onlyor read/write.

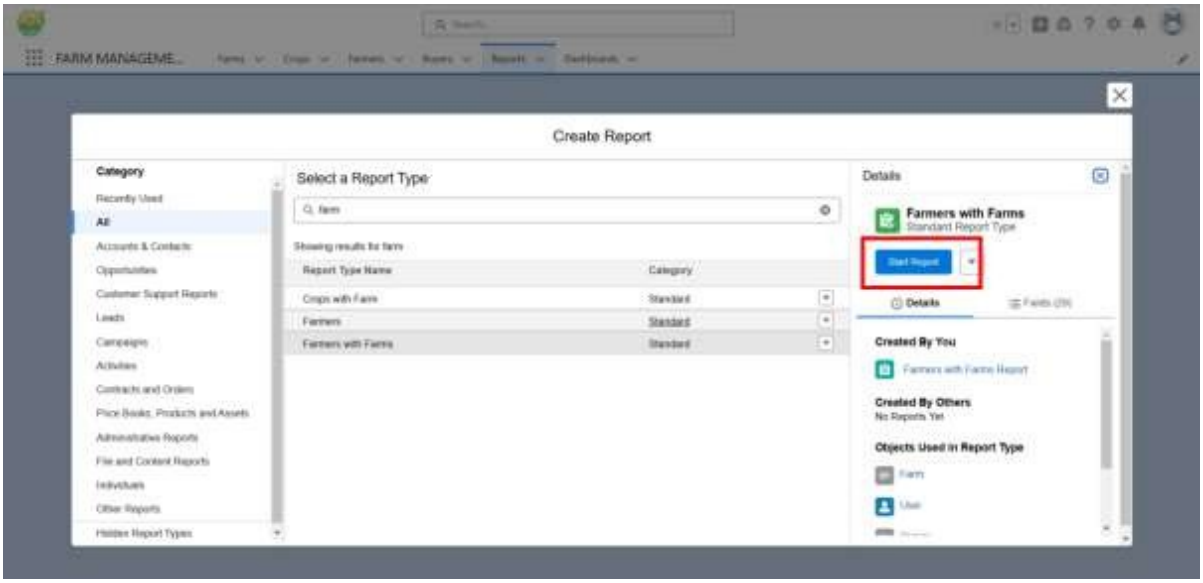
Activity1:

Createareport:

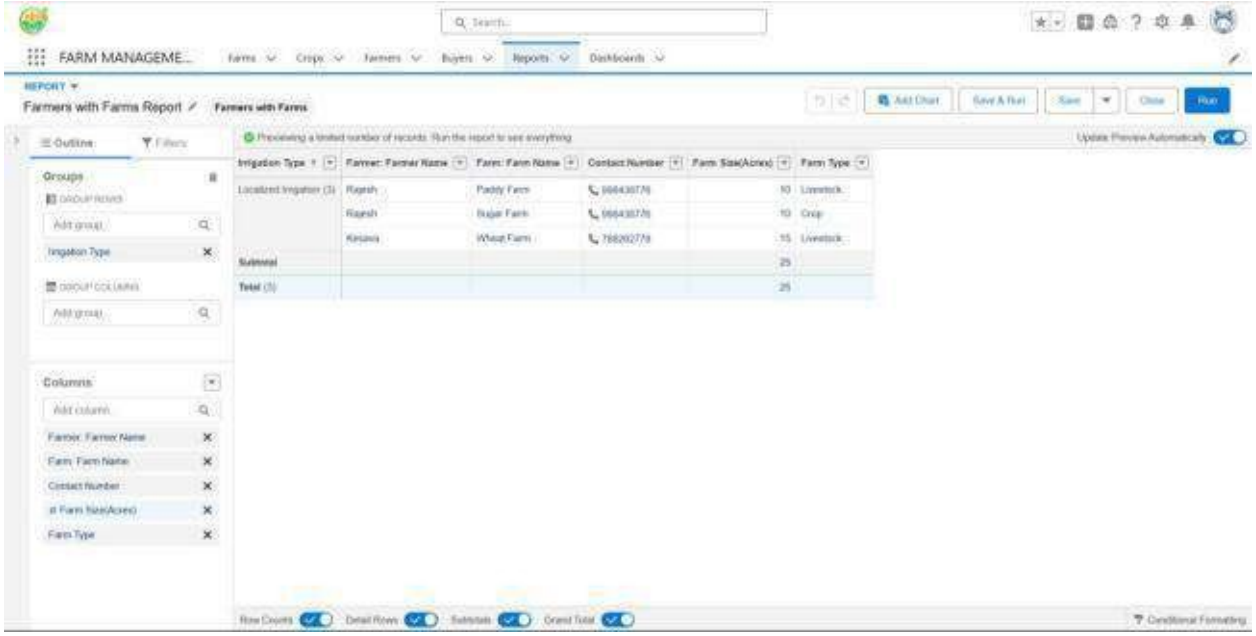
Createareportthatdisplaysratingoftheaccountandwhichhastypeandaccountname.

1) Clickonapplaunchersearchforreports.

2) ClickonthenewreportandselectthecategoryhasFarmerwithFarms.



3) Inthedetailssectionselecttheoptionstartreport.
4) InOutlineClickonColumnSelectWhichFieldyouwanttodisplayintheReport.
5) ClickGroup,SelectWhichFieldsYouWanttoGroup.
6) Savethereportbygivinglabelname(FarmerwithFarms)andsavethefolderasa public folder and save the report.

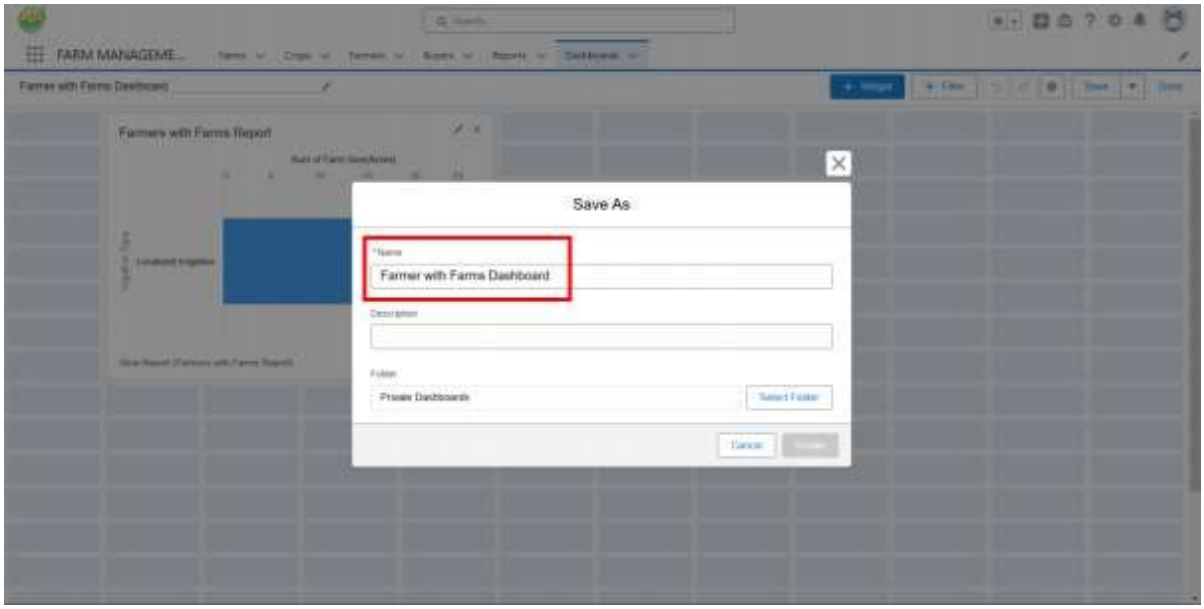


Activity2:
CreateaReportusingtheObjectsCrops,Buyers.FollowthestepsfromaboveActivity.

Milestone12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports andshow a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. Thedashboardscancontaincharts,graphsand Tabular data.

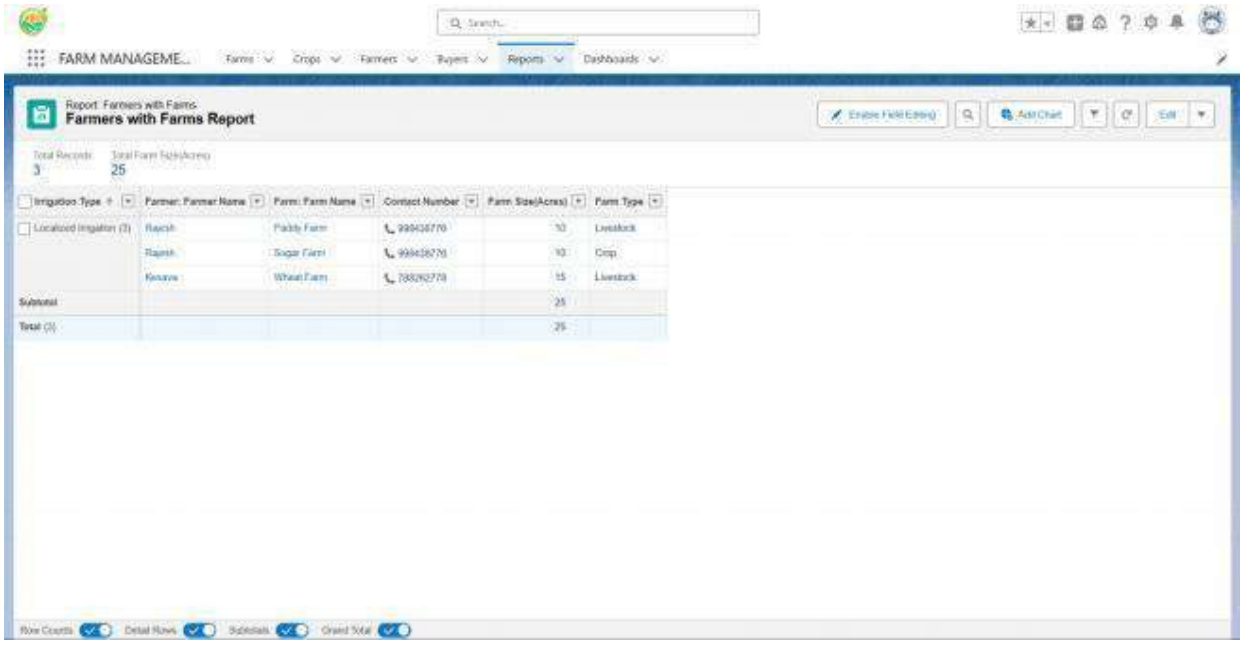
Activity1:
CreateaDashboard:
1. ClicktheDashboardtab.
2. ClickNewDashboard.
3. NamethedashboardFarmerswithfarmsandclickCreate.
4. Click+Component.
5. SelecttheFarmerswithFarmsandclickSelect.
6. SelecttheGaugechartandclick Add.
7. ClickSaveandthenDone.



Milestone-13:ViewReportandDashboard: Activity

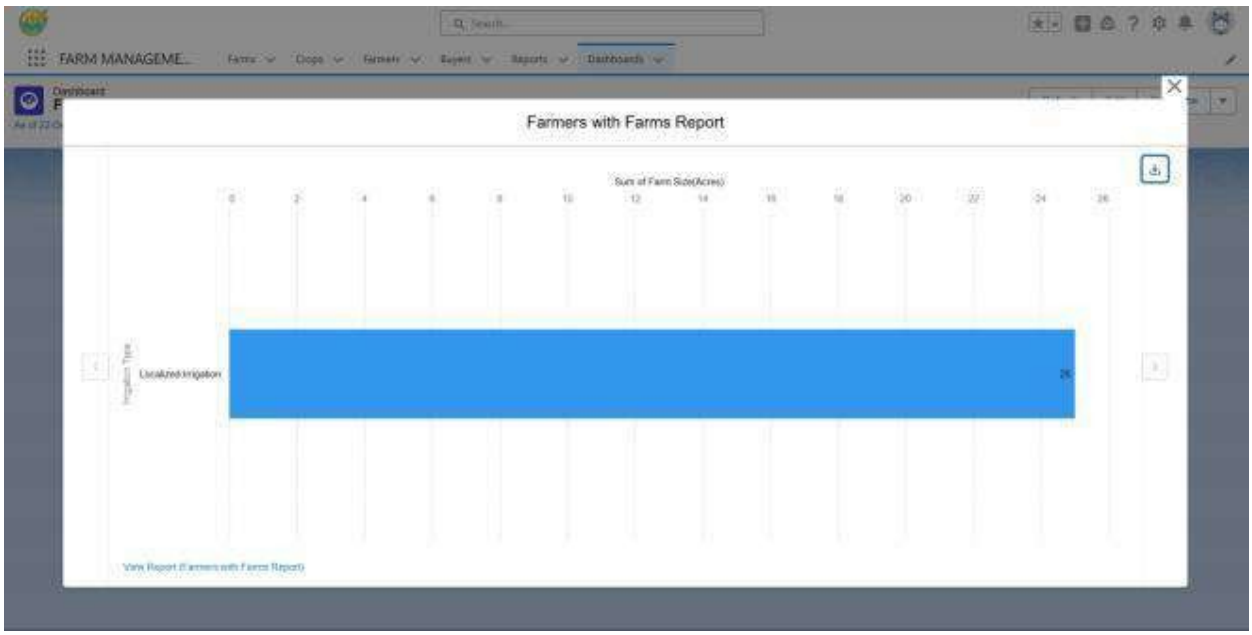
1: Report

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementSystem&clickonit.
- ClickonReports Tab.
- ClickonFarmerswithFarms&seerecords



Activity2:Dashboard

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementsystem&clickonit.
- ClickonDashboardTab.
- ClickFarmersWithfarms&seerecords.



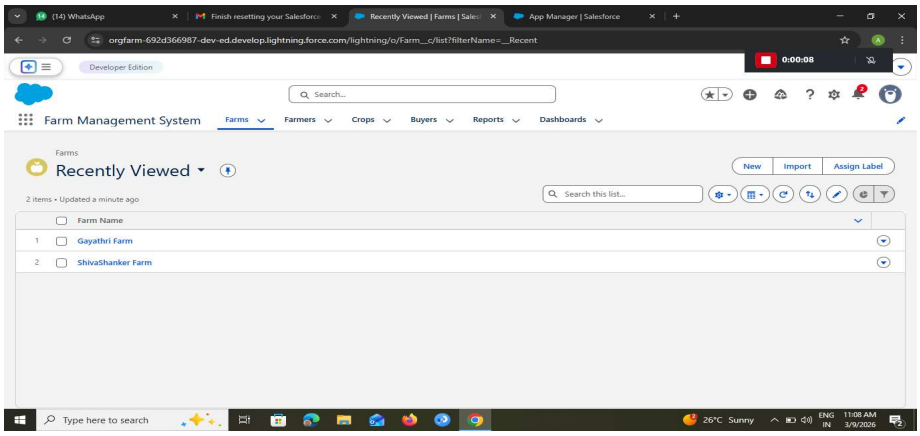
7. FUNCTIONAL&PERFORMANCETESTING

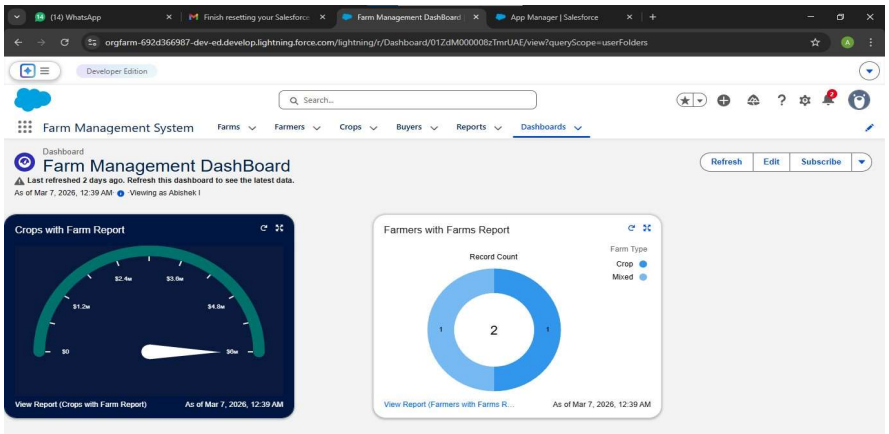
Duringthetestingphase,screenshotswerecapturedfromtheSalesforceinterfacetoverify system functionality. The following were tested:

- CreationofFarm,Crop,Farmer,andBuyerrecords
- Validationofinputfields
- Relationshipanddependencychecks
- Profileandroleaccesstesting
- Sharingruleverification
- Reportanddashboardgeneration

Performancetestingensuredsmoothsystemresponseandstabilityduringmultipleuser operations.

SCREENSHORT





p

8. ADVANTAGES& DISADVANTAGES

Advantages

- Centralizeddatasystem
- Reducedpaperwork
- Improvedaccuracy
- Bettermonitoring
- Real-timeanalytics

Disadvantages

- Requiresinternetconnection
- Salesforcelicensedependency
- Limitedofflineaccess

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURESCOPE

- Mobileapplicationintegration
- AI-basedcropprediction
- Weatherdataintegration
- IoTsensorconnectivity
- Governmentschemeintegration
- Marketpriceforecasting

11. APPENDIX

- SourceCode:SalesforceConfigurationandMetadata
- Dataset:Farm,Crop,Farmer,andBuyerRecords
- ToolsUsed:SalesforceDeveloperOrg

Thank You